METRIC

DOD-B-24505C(SH)
8 July 1980
SUPERSEDING
DOD-B-24505B(SH)
13 July 1977
(See 6.7)

MILITARY SPECIFICATION

BATTERY, STORAGE, SILVER-ZINC ALKALINE TYPE

(FOR DSRV) (METRIC)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers silver-zinc alkaline storage batteries used in the Deep Submergence Rescue Vehicle (DSRV).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-650 - Brazing Alloy, Copper, Copper-Zinc, and Copper-Phosphorus.
QQ-B-654 - Brazing Alloy, Silver.
QQ-N-290 - Nickel Plating (Electrodeposited).
QQ-S-365 - Silver Plating, Electrodeposited, General Requirements for.
PPP-B-576 - Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-300 - Chemicals, Liquid: Packaging and Packing of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6140

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation-Packaging, Methods of.
MIL-C-915 - Cable and Cord, Electrical, for Shipboard Use, General Specification for.
MIL-C-915/6 - Cable, Electrical, 600 Volts, Types SHOF, DHOF, THOF and FHOF.
MIL-C-915/58 - Cable, Electrical, 600 Volts, Type MWF.
MIL-E-917 - Electric Power Equipment, Basic Requirements.
MIL-C-5015 - Connectors, Electric, AN Type, General Specification For.
MIL-B-7883 - Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-W-8604 - Welding of Aluminum Alloys, Process for.
MIL-C-9084 - Cloth, Glass, Finished, for Resin Laminates.
MIL-R-9300 - Resin, Epoxy, Low-Pressure Laminating.
MIL-Q-9858 - Quality Program Requirements.
DOD-P-15816 - Parts, Hard Rubber for Use with Submarine Storage Batteries.
DOD-P-15817 - Parts, Synthetic Rubber, for Use with Batteries or in Battery Compartments of Submarines (Metric).
MIL-W-16878 - Wire, Electrical, Insulated High Temperature.
MIL-C-22750 - Coating, Epoxy - Polyamide.
MIL-C-22751 - Coating System, Epoxy - Polyamide, Chemical and Solvent Resistant: Process for Application of.
MIL-M-24041 - Molding and Potting Compound, Chemically Cured, Polyurethane.
MIL-P-25421 - Plastic Materials, Glass Fiber Base - Epoxy Resin, Low Pressure Laminated.
MIL-P-46120 - Plastic Molding and Extrusion Material, Polysulfone.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MS 3106 - Connector, Plug, Electric, Straight, Solder Contacts, AN Type.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
B 21 - Naval Brass Rod, Bar, and Shapes.
B 36 - Brass Plate, Sheet, Strip, and Rolled Bar.
B 121 - Leaded Brass Plate, Sheet, Strip, and Rolled Bar.
B 152 - Copper, Sheet, Strip, Plate, and Rolled Bar.
B 209 - Aluminum-Alloy Sheet and Plate.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. AGENT
National Motor Freight Classification

(Application for Copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA Traffic Dept., 1616 "P" Street, N.W., Washington, DC 20036.)

DEPARTMENT OF TRANSPORTATION (DOT)
Code of Federal Regulations - Title 49, Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article inspection. Prior to beginning production, six cells shall be examined and tested as specified in 4.3.1 (see 6.3).

3.2 General requirements.

3.2.1 Design and construction. The battery installation shall consist of two separate batteries of 76 series-connected cells. Each battery shall consist of cells, battery box and cover, intercell connectors, battery electrical connectors, fuses, cell voltage monitoring harness, venting and compensating system connections, shunt, salt water detector and compensating oil. In addition, the battery shall be capable of performing a rescue mission as defined in 3.3.1.1. Typical configuration is shown on figure 1.

3.2.2 Installation and interface. Each battery shall be installed external to the ship pressure hull. Cells shall be open to the pressure-compensating oil filling the battery box and serving as a buffer between the electrolyte and the seawater. The contractor shall be responsible for the interface between the battery and the ship. The Government will

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

make available all necessary information to the contractor upon request during the design review meetings (see 3.10). In case of conflict with this specification, the contractor shall be responsible for requesting the necessary changes.

3.3 Electrical requirements.

3.3.1 Electrical characteristics. The characteristics of each 76-cell battery, based on an electrolyte temperature and an ambient temperature as low as 0°C ± 3°C at the start of the discharge and a constant pressure as high as 25 megapascals (MPa) shall be not less than that shown in table I.

TABLE I. Electrical characteristics.

Discharge rate amperes (A) 1/	Capacity (Ah) 1/	Battery minimum voltage (V)	Minimum average voltage per cell 2/ (V)	Minimum (cut-off) cell voltage (V)
300	600	100	1.40	1.25
170	680	100	1.45	1.31
140	700	100	1.47	1.34

1/ The capacity and the voltages at any discharge rate below 300 A but not shown on the table shall be respectively at least equal to the capacity and the voltages at the next higher rate shown.

2/ The average voltage is defined as the average of voltage recordings taken at equal intervals, starting with the first recording (taken between 1/2 minute and 1 minute after applying a constant rate) and including the last recording. The interval of recording shall be not greater than h/5 hours where h is the time in hours of the rated capacity of the test. The time to the last recording shall be not greater than the rated time.

3.3.1.1 Simulated rescue mission. The simulated rescue mission requirement is defined as a series of 10 partial cycles in succession without the occurrence of grounds, shorted cells, capacity loss or other factors limiting the battery performance. The partial cycle is defined as a discharge up to 400 Ah at any rate of table I followed by a recharge at 100 A to the cut-off voltage recommended by the service manual. The battery shall be capable of meeting this requirement during the first 12 months of its life.

3.3.2 Life. The battery, when installed, shall be capable of delivering a total of 40,000 Ah of discharge over a period of 18 months. Except as required by 4.6.3.2, the capacity shall be at least 500 Ah when discharged at 100 A to a voltage cutoff of 1.35 volts.

3.3.3 Charge.

3.3.3.1 Full charge definition. Full charge is defined as an Ah input at least equal to 90 percent of the Ah output obtained in the prior discharge.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

3.3.3.2 Charge acceptance. The battery or single cells shall be capable of accepting, in 16 hours or less, a full charge as defined in 3.3.3.1 under any one or a combination of the following conditions:

(a) After a discharge at any of the rates and for the corresponding cut-off voltages shown in table I.
(b) Using any charging rate or combination of rates up to 100 A as selected by the Naval Sea Systems Command (NAVSEA).
(c) An electrolyte starting temperature as low as 0°C ± 3°C.
(d) An ambient pressure as high as 7,000 kilopascals (kPa).

3.3.3.3 Individual cell charging. Without the removal of the cell from the battery or the destruction of the insulation of the terminals or the connectors, each cell shall be provided with means of charging or discharging it individually at a current up to 20 A, if desired, for balancing or trouble shooting purposes.

3.3.4 Cell terminals. The temperature at the top of the positive terminal shall not exceed 60°C at any time during a discharge at 300 A when tested as specified in 4.6.3.7.

3.3.5 Hydrogen evolution. Hydrogen evolution from any charged cell shall not exceed 20 milliliters (mL) per hour average over a 24-hour period when the cell is maintained on open circuit 25°C ± 3°C.

3.3.6 Internal cell resistance. The internal cell resistance of any cell shall not deviate more than 20 percent from the average value when measured as specified in 4.6.4.2.

3.4 Physical requirements.

3.4.1 Battery box dimensions. Overall dimensions of each battery box and cover shall be as shown on figure 1.

3.4.1.1 Battery box materials. The battery box and cover shall be glass reinforced epoxy using glass cloth in accordance with MIL-C-9084, type VIIIB, class 2 and epoxy resin in accordance with MIL-R-9300, type I, grade D, form A or B. The fabricated laminate shall meet the requirements of MIL-P-25421, type I, class 2. Samples for testing may be taken from trim material or a concurrent flat panel may be fabricated along side the box and cover using the same procedure as that used on the box and cover. The hanger shall be constructed of aluminum alloy in accordance with ASTM B 209, alloy 6061, welded in accordance with MIL-W-8604 and finished in accordance with MIL-C-22750 and MIL-C-22751, white No. 7785 in accordance with FED-STD-595. Mounting holes in the hanger shall be drilled by the contractor using the Government-loaned tool (see 3.8).

3.4.2 Mass. The mass of each battery box and cell shall not exceed 1185 kg and 13 kg respectively. The mass of the battery box is complete with cells, electrolyte, compensating oil, scanners, shunt, fuses, electrical cables, connector plate and battery case and cover with compensation and vent fittings. The mass of each cell is complete with electrolyte (and not including oil).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

3.4.3 Cell jar. Cell jars shall be molded out of polysulfone in accordance with MIL-P-46120 using the Government-owned mold (see 3.8) and be inspected in accordance with 4.6.1.1. Typical configuration is shown on figure 3. Cell jars shall be supported by neoprene pads 3 mm thick with 6 mm clearance around all edges.

3.4.4 Cell cover. Cell covers shall be molded out of the same material as the jar material using the Government-owned mold (see 3.8) specified in the contract or order (see 6.2.1) and be inspected in accordance with 4.6.1.1. Typical configuration is shown on figure 4.

3.4.5 Cover-to-jar sealing. The cover-to-jar sealing shall be 3M compound 3520 structural adhesive or equal.

3.4.6 Electrode pack.

3.4.6.1 Grids. Silver shall be used for the positive and negative grids supporting the active materials. The equivalent thickness of each grid (positive or negative) when reduced to a solid sheet of the same mass and same apparent electrode area, shall be at least 0.05 mm.

3.4.6.2 Tabs. Silver shall be used for the positive and negative electrode tabs attached to the terminal base. The tab area below the top of the grid shall be at least 90 percent hot-forge bonded to the grid.

3.4.6.3 Silver active material. Silver active material amount shall be at least a nominal 3.65 grams per Ah, based on the 700-Ah capacity.

3.4.6.4 Electrode construction. The positive electrodes shall have smooth edges, and corners rounded to a radius of at least 6 mm. The negative electrode grids shall have smooth edges, and corners rounded to a radius of at least 6 mm.

3.4.6.5 Negative electrode. The use of mercury in the negative electrode is prohibited. A mercury substitute designated as NAVSEC I (1 percent lead oxide and 2 percent cadmium oxide) shall be used.

3.4.6.6 Separator system. The separator system shall be as follows:

(a) The separator system shall consist, from positive electrode to negative electrode, of a positive absorbent material layer, cellophane-type main separator layers and a negative absorbent material layer.
(b) The positive and negative absorbent material shall be at least 0.08 to 0.17 mm thick. The material shall be a layer of nonwoven polyamide material from Pellon Corporation, or equal. The top shall be 5 to 6 mm over the top of the electrodes.
(c) The main separator shall be Dupont Cellophane PUDO-193 (battery grade) (as is or treated) alone or in combination with Union Carbide Fibrous Sausage Casing (battery grade), or equal. The overall wet thickness of the combined layers of this material when swollen in the specified electrolyte, shall be at least 0.45 mm. The main separator height over the top of the electrodes shall be at least 50 mm.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

(d) The main separator assembly method shall consist of bagging or wrapping the positive or negative electrodes separately. If bagging or wrapping of a pair of electrodes is used, it shall be done on the positive electrode. If wrapping is used, the overlap of the last layer shall be at least 15 mm.

(e) Epoxy or cement sealing of the bottom of separator shall not be used.

3.4.7 Terminals. That part of the terminal mating with the plug-in type intercell connector (see 3.4.8.2) shall be made of Copper Alloy No. 155 or equal. Other current carrying terminal parts shall be made of solid copper or copper alloys in accordance with ASTM B 21, B 36, B 121, or B 152. Terminal parts shall be silver-plated in accordance with QQ-S-365, type III, grade B, with a nickel undercoat in accordance with QQ-N-290, class 1, grade F, form SB. Assembly of the terminals shall be by mechanical attachment or brazing. Soldering shall not be used. Brazing shall be in accordance with MIL-B-7883, grade B, using brazing alloys in accordance with QQ-B-650 or QQ-B-654. Terminal posts shall be symmetrically placed with respect to an axis through the center of the cell top. Terminal posts shall be provided for each polarity and shall be a part of a single, silver-plated copper strap to which the plate tabs will be attached. Terminal posts shall be circular in cross section.

3.4.8 Intercell connectors. Intercell connectors shall be in accordance with the contract or order. The completed plug-in intercell connector design shall be confirmed during the first design review. Provisions shall be made to measure the voltage drop between the terminal and the intercell connector.

3.4.9 Rubber parts. Soft rubber parts shall conform to DOD-P-15817. Hard rubber parts shall conform to DOD-P-15816. The material of O-rings, gaskets, and seals shall be neoprene or ethylene-propylene rubber, Hypalon or equal, except when used in the electrolyte entrainment eliminators (see 3.4.10.4), it shall be Hypalon.

3.4.10 Electrolyte.

3.4.10.1 Composition. The electrolyte used for filling the cells shall be an aqueous KOH solution of 45 percent plus or minus 1 percent KOH by mass with no additives. The maximum concentration of impurities allowable in the electrolyte prior to filling shall be as listed in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

TABLE II. Maximum concentration of impurities in electrolyte.

Impurities	Maximum concentration (percent based on mass)
Potassium carbonate	0.50 percent
Chloride (as Cl)	0.005 percent
Sulfate (as SO_4)	0.002 percent
Nitrogen compounds (as N)	0.001 percent
Ammonium hydroxide precipitate	0.01 percent
Phosphate (as PO_4)	5 p/m
Iron	5 p/m
Sodium	0.02 percent
Other metals total combined	2 p/m
Reducing material	See note

Note: Reducing material shall be determined at 75°C ± 5°C in an acid medium by a maximum of 1.5 mL of 0.001 normal potassium permanganate solution reacting completely with reducing material contained in a 10 mL sample of KOH solution.

3.4.10.2 Level indication. The manufacturer shall provide a device by which the electrolyte level may be determined for the purpose of electrolyte adjustment.

3.4.10.3 Level. At no time during operation shall the electrolyte level variation be more than 50 mm from the highest level to the lowest level.

3.4.10.4 Entrainment eliminator. Each cell shall be fitted with an electrolyte entrainment eliminator which shall be attached to the vent hole in the cell cover. The eliminator shall prevent carry-over of electrolyte from the cell to the oil-filled battery box under the conditions of battery usage and operation. Baffles may be used under the vent hole to assist in entrainment elimination. The entrainment eliminator shall meet the test specified in 4.6.3.8.

3.4.11 Cooling. No internal cell cooling device requiring the circulation of coolant is permitted. Cooling shall only be by natural convection of the dielectric oil filling the spaces between the cells.

3.4.12 Gas-tightness. Cells shall be capable of withstanding 35 kPa differential air pressure for 5 minutes without loss of pressure.

3.4.13 Cell identification. Cell identification shall be as specified in 3.4.13.1 and 3.4.13.2.

3.4.13.1 Cell markers. One set of individual cell markers shall be furnished per battery. Cell markers shall be consecutively numbered from 1 through 76, preceded by the battery number given by NAVSEA (for example 4-25 for battery number and cell number). The cell markers shall be attached to the tops of the cells.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

3.4.13.2 Serial numbers. The manufacturer's serial numbers shall be visible from the top of the installed cells and shall be indelibly marked. The serial number shall consist of five digits: the first two digits shall represent the year of manufacture and the last three digits shall be serially assigned by the manufacturer.

3.4.14 Battery definition. When only cells are specified (see 6.2.1), the batteries shall consist only of cells.

3.4.15 Battery assembly. When specified (see 6.2.1), the battery assembly shall be completed as specified herein and all inspection on cables of 4.4.2 shall apply.

3.4.15.1 Metal insulation. Possible sources of electrical potential inside the battery box, including, but not limited to, cell terminals, intercell and battery connectors, fuses, shunt, metal screws or bolts, and connection to compensating systems, shall be insulated in such a manner that no electrolyte or seawater can come in contact with the metal of these items at any time to cause grounds during the life of the battery. Insulation material applied to intercell connectors shall not require more than 24 hours of removal, application and cure. Potting and encapsulating procedure shall comply with MIL-M-24041 and appendix A herein.

3.4.15.2 Cell voltage scanner. Each battery box shall contain a cell voltage scanner (see 3.8) furnished by the Government for use with a monitoring system. The scanner mass shall be not more than 22 kilograms (kg) which shall be included in total battery mass.

3.4.15.3 Fuses. A 700 A fuse shall be furnished between each power connector and the battery. Each individual cell voltage (ICV) wire shall have a removable 10 A fuse connected in line, located as close as possible to the cell terminal.

3.4.15.4 ICV wiring and signal cables. Wiring for individual cell voltage leads shall be in accordance with MIL-W-16878, 16-gage wire, except that silver plated conductor is not required. The wiring shall be provided with connections at both ends of the harness. Insulation material shall be resistant to potassium hydroxide (KOH), seawater, and dielectric oil (see 3.4.18), and capable of withstanding the specified temperature range (see 3.5.1). The wiring arrangement shall be as shown on figure 2. Signal cables shall be in accordance with MIL-C-915 and MIL-C-915/58, type MFW-19. Hypalon compound 70H23 may be added as an additonal layer of insulation on ICV wiring and signal cables. ICV wiring and signal cables shall be certified as passing the cable inspection of 4.4.2.

3.4.15.5 Power cables. Power cables shall be in accordance with MIL-C-915 and MIL-C-915/6, type SHOF-150. Cables shall be certified as passing the cable inspection of 4.4.2.

3.4.15.6 ICV wiring and signal cable connectors. ICV wiring and signal cable connectors shall be in accordance with MIL-C-5015 and MS 3106.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

3.4.15.7 Battery box seal. The battery box cover shall be fastened to the box by means which will allow access as necessary for repair and maintenance. The seal between the box and cover shall be gas and liquid tight. The assembled box and cover shall be capable of withstanding an internal air pressure differential of 14 kPa for 5 minutes without loss of pressure.

3.4.15.8 Battery shunt. Each battery box shall contain a mounted and insulated shunt rated at 5 A per millivolt with a minimum capacity of 300 A.

3.4.15.9 Salt water indicating probe. Each battery box shall be fitted with a salt water indicator probe located 80 to 100 mm above the inside bottom of the box inside the scanner well. The probe shall have appropriate lead wires to a connector in the battery box cover.

3.4.15.10 Battery connectors. Connectors shall be in accordance with figures 1 and 2. The receptacle plate shall be removable from the battery box (unpluggable). No bare metal shall be exposed. There shall be three connectors: one power connector for each polarity and one signal connector. These three connectors shall be located in the connector plate in the battery box cover over the scanner well as shown in figure 1.

3.4.16 Battery box connections.

3.4.16.1 Mechanical. The compensator system fitting and the two venting system fittings shall be provided and located on the battery box cover as shown on figure 1. The compensator fitting, also used to pump oil from the battery before removing the cover, shall be provided with a flexible tube extending from the fitting down into the scanner well to within 25 mm of the battery box bottom. The compensator fitting shall mate with the quick-disconnect fitting in use on the ship. Interface details regarding the quick-disconnect fitting configuration shall be presented during the design review meeting (see 3.10). The venting system fittings shall consist of a relief vent valve to allow gas venting from the battery under internal excess differential pressure of 20 kPa $\pm$ 2 kPa, and a manual vent valve to be opened only to allow air to enter the box when oil is being strained or pumped out.

3.4.16.2 Electrical. Electrical connections shall be made through the battery box cover as shown on figures 1 and 2. The portion of the battery box cover through which electrical connections are made shall be separate from the main body of the cover. The box cover shall be capable of being removed without having excess cable length or requiring disconnection of electrical connections.

3.4.17 Dielectric oil. EXXON MARCOL oil 70, or equal shall be used in the cells, battery box, and compensating system. Dielectric oil shall contain an "oil blue N" dye (methyl-amino-butylamino anthraquinone) to facilitate visual distinction between oil and electrolyte.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

3.4.18 Lifting brackets. The battery box shall be provided with lifting and mounting brackets attached to the four corners of the box as shown typically on figure 1. Brackets shall be bolted to the battery box and welded to a hanger assembly which shall fit under the battery box top edge flange on all four sides. Metallic parts of brackets inside the battery box shall be insulated. Detailed interface information shall be presented in design review meetings (see 3.10) to insure proper mating for installation of the battery in the ship.

3.5 Environmental requirements.

3.5.1 Temperature. The battery shall be capable of withstanding temperatures up to 70°C for 2 hours at any high rate discharge while the range for normal operation is minus 2°C to 32°C.

3.5.2 Pressure. The battery shall be capable of continuous operation throughout the specified life cycle with dielectric oil filling the cells and the battery box. It shall be capable of withstanding in this condition a compensating pressure as high as 25 MPa when submerged in seawater.

3.5.3 Angle of operation. The battery shall be capable of continuous operation at an angle of 45 degrees from the vertical without physical or electrical damage or electrolyte spillage from any cell. Dielectric oil shall not come in contact with the electrodes when the cells are tilted 45 degrees in any direction.

3.6 Cells, repair parts, and accessories.

3.6.1 Cells. The total number of cells to be delivered per ship shall comply with the following:

(a) Two batteries, each consisting of 76 individual cells, for installation in the ship shall be shipped as specified (see 6.2.1).
(b) Eight spare cells, for possible replacement of failed cells in the installed batteries. Cells shall be kept dry and shipped as specified (see 6.2.1).

3.6.2 Repair parts. Repair parts are all parts needed for repair, replacement, maintenance, and installation. Repair parts shall be provided as specified (see 6.2.1) and shall include, the following for each set of 2 batteries:

(a) Thirty entrainment eliminators.
(b) Requisite electrolyte for all dry cell filling (to be provided in separate polyethylene carboys of 3 to 4 liters).
(c) Two hundred 10-A fuses for ICV wiring.
(d) Two level indication devices.
(e) Two electrolyte adjustment devices.
(f) Eight cell marker blanks.
(g) Materials or parts deemed necessary by the manufcturer for repair, replacement, maintenance, and installation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

(h) Two plug-in intercell connectors.
(i) Two plug-in cell jumper cables.

3.6.3 Accessories. The following accessories (parts for one cell) shall be provided as specified (see 6.2.1):

(a) Jar.
(b) Cover.

3.7 Engineering services. Engineering services of the contractor's engineer shall be provided as specified (see 6.2.1) upon NAVSEA's request in order to:

(a) Assist the shipyard in preparing the battery for installation and installing it.
(b) Assist the ship's force in becoming familiar with the battery.
(c) Take corrective action in case of unexpected problems.
(d) Examine the battery condition when requested.
(e) Participate in technical meetings with NAVSEA or the ship's force with regard to the battery and in locations other than the manufacturer's plant. Design review meetings shall not be considered a provision of engineering services.

The maximum number of mandays allowed for the entire contract or life of one set of batteries shall be as specified (see 6.2.1). Field engineering travel reports shall be prepared in accordance with the data ordering document included in the contract or order (see 6.2.2).

3.8 Government-furnished items. The Government will make available the following items:

(a) Government-owned molds:
(1) Cell jar (see 3.4.3).
(2) Cell cover, 4 terminals (see 3.4.4).
(3) Cell cover, 2 terminals (see 3.4.4).
(b) Government-furnished equipment:
(1) Individual cell voltage scanner (see 3.4.15.2).
(2) Tool No. 2653479-501 17 EX-2 (see 3.4.1.1).

3.9 Technical data. The contractor shall prepare technical data in accordance with the data ordering documents included in the contract or order (see 6.2.2) and as specified in 3.9.1 through 3.9.4. The contractor shall use the International System of Units (SI) in all technical data except for threaded hardware designation where the Unified Thread Standard shall be used.

3.9.1 Drawings. The contractor shall prepare engineering drawings (including documents referenced therein) for review by NAVSEA in accordance with the data ordering document included in the contract or order (see 6.2.2) and as described in 3.9.1.1 and 3.9.1.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

3.9.1.1 Nonproprietary documents. Nonproprietary documents shall not include or refer to proprietary information or refer to any other document not supplied. Nonproprietary documents shall include the following:

(a) Cell jar.
(b) Cell cover.
(c) Electrolyte level indicator device.
(d) Entrainment eliminator.
(e) Terminal.
(f) Sketch showing relative positions of tops of plates and separators with respect to cover underside.
(g) Terminal-cover assembly.
(h) Cell control drawing.
(i) Cell pack assembly.
(j) Electrode and tab dimensions.
(k) Characteristics of cell electrode pack components: number of electrodes; mass of active material per electrode; type, number and thickness of absorbent materials and separators between electrodes; grid type used in electrodes and its mass in g/cm^2.
(l) Battery box.
(m) Battery cover.
(n) Battery cell layout.
(o) Battery wiring and cabling.
(p) Battery connections.
(q) Data list (list of all drawings that are used in the manufacture of the cells and the battery).

3.9.1.2 Proprietary documents. Proprietary engineering drawings (including documents referenced therein) shall be available to NAVSEA for review at any time.

3.9.2 Service manual. The contractor shall prepare a service manual in accordance with the data ordering document included in the contract or order (see 6.2.2) and appendix B. The service manual shall include a curve and data table showing charging rate efficiencies for output versus input for charging rates up to 200 A. The service manual shall have a NAVSEA number which will be supplied by NAVSEA.

3.9.3 Milestone plan. The contractor shall provide and maintain a milestone plan and prepare a report in accordance with the data ordering documents included in the contract or order (see 6.2.2).

3.9.4 Engineering change proposals. The contractor shall prepare engineering change proposals in accordance with the data ordering document included in the contract or order (see 6.2.2).

3.10 Design review. Design review meetings with Government representative shall be held prior to the release of any drawings to contractors or production or to finalization of the service manual. At least three meetings (two at NAVSEA) are required: one prior to design to establish a design concepts agreement and provide any interface information, one to review drawings and design, and one to discuss the service manual.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contactor may use his own or any other facilities suitable for the performance of the inspection requriements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality program. The contractor shall provide and maintain a quality program acceptable to the Government for supplies and services covered by this specification. The quality program shall be in accordance with MIL-Q-9858.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3.1).
(b) Quality conformance inspection (see 4.4).

4.3 Inspection.

4.3.1 First article inspection. Six cells for the entire contract shall be subjected to the first article inspection which shall consist of the examination specified in 4.6.2.1 and the tests specified in 4.6.2.2 and 4.6.3 through 4.6.3.9. Within 15 days after completion of the first article inspection, the cells shall be shipped with all necessary parts and instructions to a Government laboratory (see 4.7), for life testing (see 4.8).

4.3.2 Sampling for quality conformance inspection. A sample of four cells shall be randomly selected from each production lot, representing two 76-cell batteries plus the 16 spare cells, for quality conformance inspection (see 4.4) and life testing (see 4.8).

4.4 Quality conformance inspection.

4.4.1 Cell inspection. The entire production lot shall be subjected to the inspection of 4.6.2. The sample cells shall be filled, formed, and subjected to the inspections specified in 4.6.4. After completion of the quality conformance inspection, the sample cells shall be shipped to a Government laboratory (see 4.7), for life testing (see 4.8).

4.4.2 Cable inspection. Cable examination and tests (see 4.4.2.1) shall be conducted prior to and after pressurization cycling (see 4.4.2.2). Recorded data shall be reported in the cable certification report (see 4.11.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

4.4.2.1 Examination and tests. The following cable examination and tests shall be conducted:

(a) Visual examination. Visually examine the cable to insure it is free of visible physical damage. Record the result of the visual examination.
(b) Continuity test. Using an ohmmeter, verify the continuity of each conductor path. Record the result including the cable type and specific conductor. The resistance per conductor shall be less than one ohm.
(c) Insulation resistance test. Using a 500 V direct current (d.c.) megger (megohmmeter), measure the insulation resistance between each wire and all other wires and outer insulation of the cable. Record the insulation resistance. The insulation resistance shall be not less than 100 megohms.

4.4.2.2 Cable pressurization cycling. Prior to pressurization, all cables shall be sealed against moisture and pressure using potted end seals. Potting shall be in accordance with appendix A. Cables shall then be immersed in water and subjected to a pressure of 80 MPa, held for 10 minutes, then depressurized. This shall be repeated for a total of nine cycles. The tenth and final cycle shall be to the pressure above, held for 1 hour before depressurization. For each pressure cycle, the following data shall be recorded:

(a) Maximum hold pressure.
(b) Start, stop, and hold times.
(c) Temperature.

4.4.3 Battery or cell inspection. When notified (see 4.6.4.4), the contractor shall subject the batteries or cells to the tests specified in 4.6.4.

4.5 Battery acceptance.

4.5.1 Acceptance in the dry condition. Prior to acceptance by the cognizant Defense Contract Administration Service Management Area (DCASMA), the technical data specified in 3.9.1, 3.9.2, 4.10, 4.11.1, 4.11.2, 4.11.4 and 4.11.5 shall have been delivered and accepted by NAVSEA, the cells in the dry condition shall have passed the examination and tests specified in 4.6.2, and the sample cells shall have passed the tests specified in 4.6.4. After acceptance of the dry battery by the cognizant DCASMA, the contractor shall be responsible for the storage of the dry battery in proper conditions and its subsequent physical and electrical performance after the battery is filled and formed, and until accepted again in accordance with 4.5.2.

4.5.2 Acceptance in the wet condition. Upon written notification by NAVSEA within 2 years after acceptance in the dry condition, the contractor shall be responsible for filling and forming the battery as required by 4.6.4. After establishment of satisfactory performance in accordance with 4.6.4, the cognizant DCASMA, will then accept the battery for delivery and shipment as directed by NAVSEA.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

4.6 Inspection methods.

4.6.1 Inspection of parts. Parts used in the assembly of the cells shall be examined on a sample basis in accordance with MIL-STD-105, inspection level II acceptance quality level (AQL) equals 1.0 percent defective for conformance to the requirements of this specification and to the manufacturer's quality control requirements.

4.6.1.1 Acceptance criteria for jars and covers. Molded jars and covers shall be inspected to determine conformance with the following requirements:

(a) Color. Material shall be transparent and natural (light amber) in color.
(b) Surface defects. Parts shall be free of blisters, gas marks, cracks, or wrinkles and reasonably free from small defects such as scratches, dents, and heat marks.
(c) Sub-surface defects. Parts shall be free of blisters, gas marks, voids, and other defects detrimental to its appearance or serviceability.
(d) Dimensions and tolerances. Dimensions and tolerances shall be as specified in applicable drawings.
(e) Core shift. Defects resulting from core shift are not acceptable.

4.6.1.2 Acceptance criteria for battery box and cover. The battery box and cover shall be inspected to determine conformance with the requirements of 3.4.1 and 3.4.1.1, and with the following requirements:

(a) There shall be no open voids, pits or cracks.
(b) There shall be no areas of delamination.
(c) There shall be no protruding surface fibers.
(d) There shall be no surface crazing.
(e) Exposed edges of laminate which have been cut or sanded shall be sealed with a coat of general purpose resin.

4.6.2 Inspection of all cells in the dry condition.

4.6.2.1 Examination. Cells (first article, sample, and production lot) shall be examined for workmanship and for conformance with 3.3.3.3, 3.4.1, 3.4.13 and 3.4.15.1. Any defective cell shall be rejected.

4.6.2.2 Gas-tightness. Each cell shall be tested to determine conformance with 3.4.12. Any defective cell shall be rejected.

4.6.3 Inspection of first article cells in the wet condition. Except where otherwise specified, the temperature of the electrolyte at the start of each test shall be within 16°C to 32°C.

4.6.3.1 Mass. The first article cells shall be weighed after formation. The mass shall meet the requirements specified in 3.4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

4.6.3.2 Capacity and voltage determination. After formation in accordance with the manufacturer's instructions and the test specified in 4.6.4.3, each cell shall be cycled at room temperature as follows:

(a) Cycle 1. Each cell shall be charged as recommended in the service manual, then placed in an oil-filled vessel, pressurized to 25 MPa. Each cell shall then be discharged at 300 A to the corresponding cut-off voltage.
(b) Cycle 2. Each cell shall be depressurized at the rate of 700 kPa per minute down to atmospheric pressure and then recharged. Each cell shall then be pressurized to 21 MPa. The cell shall then be discharged at 150 A to the corresponding cut-off voltage. The cell shall be depressurized at the rate of 700 kPa per minute down to atmospheric pressure.
(c) Cycle 3. Each cell shall then be recharged at 100 A to the recommended cut-off voltage then immediately pressurized to 25 MPa then discharged at 200 A for 2 hours. The cell shall then be depressurized at the rate of 700 kPa per minute.

Data on voltage, current, and electrolyte temperature shall be monitored and recorded. The results of the capacity discharge tests at each rate shall be averaged and the estimated standard deviation calculated according to the following formula:

$$s = \left(\frac{\sum_{1}^{n} (x_i - \bar{x})^2}{n-1} \right)^{1/2}$$

Where:

s = Estimated standard deviation.
n = Number of sample cells.
$\bar{x}$ = Average of results of test.
x_i = Result of individual sample cell.

The capacity and voltage of each cell shall meet the requirements of 3.3.1 and the average percent of rated capacity for all cells shall be equal to or greater than 100 percent of rated capacity plus 3.1 s.

4.6.3.3 Charge acceptance and efficiency. After completion of the cycling tests of 4.6.3.2, each cell shall be tested for charge acceptance and efficiency. The requirement of 3.3.3.2 shall be demonstrated only for rates up to and including 100 A on the tests specified herein. Rates higher than 100 A are to be used for information and record purposes for incorporation in the service manual. Starting with the discharged cells

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

at an electrolyte temperature and an ambient temperature of 25°C ± 5°C, each cell shall be submitted to a sequence of six cycles. Discharges shall be at 150 A to the cut-off voltage. The successive charges to the corresponding cut-off voltages shall be made respectively at 40 A, 60 A, 100 A, 150 A, and 200 A. The sixth cycle recharge rate shall be 100 A to the cut-off voltage, followed by 30 A to the corresponding cut-off voltage. The following data shall be recorded and reported for each cycle: capacity input, capacity output, and efficiency factor (output versus input). This information shall be included in the service manual (see 3.9.2).

4.6.3.4 Hydrogen evolution. Each cell shall be tested for hydrogen evolution on open circuit stand to determine conformance with 3.3.5. The open circuit voltage period shall start after the end of a charge conducted in accordance with the manufacturer's service manual. Gas collection shall commence at the beginning of the open circuit period and quantities shall be recorded periodically until completion of the test.

4.6.3.5 Angle of operation. Each cell, within 1 hour after ending any charge, shall be tilted in any direction 45 degrees from the vertical to determine conformance with 3.5.3. The cells shall be tilted, for information only, to 60 degrees from the vertical or to an angle less than 60 degrees where spill occurs, to determine the maximum inclination capability of the manufacturer's design. The data shall be recorded.

4.6.3.6 Electrolyte level indication. The method of determination of the actual electrolyte level (see 3.4.10.2 and 3.4.10.3) shall be demonstrated on actual cells for various levels which occur during charged, discharged, and open circuit states or which may occur during abnormal conditions (too high or too low).

4.6.3.7 Terminal temperature. Cells shall be connected as a battery and insulated on four sides so as to approximate an adiabatic condition as closely as possible. The temperatures of the positive terminal and electrolyte shall be measured by thermocouples or thermistors and recorded at 15-minute intervals during a 300 A rate discharge with an electrolyte starting temperature and an ambient temperature of 30°C ± 2°C. The cells shall meet the requirement specified in 3.3.4.

4.6.3.8 Electrolyte entrainment. The effectiveness of the electrolyte entrainment eliminator (see 3.4.10.4) shall be demonstrated during the tests of 4.6.3.2. There shall be no visual evidence of electrolyte carry-over out of any cell during the entire sequence of tests of 4.6.3.2.

4.6.3.9 Simulated rescue mission. In order to demonstrate conformance with the requirements of 3.3.1.1, the cells shall be charged as recommended and submitted to ten continuous cycles, consisting of a discharge at 200 A with a first discharge starting electrolyte temperature of 16°C to 32°C and under a differential pressure of 25 MPa followed by a recharge at 100 A at atmospheric pressure to the cut-off voltage recommended by the contractor. The first nine discharges shall be run for 2 hours and shall deliver 400 Ah over the cut-off voltage. The tenth discharge shall be continued to the cut-off voltage and shall deliver at least 400 Ah. The data on capacity and end voltage for each discharge shall be recorded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

4.6.3.10 Failure procedure. In case of failure, the tests of 4.6.3 may be repeated once. In case of subsequent failure, the first article cells which fail to meet individually the tests of 4.6.3 shall be rejected and may not be offered again for testing unless a proposal for the correction of the deficiency and retest of new first article cells has been submitted and reviewed by NAVSEA.

4.6.4 Inspection of wet sample cells and wet battery.

4.6.4.1 Cell formation. Sample and battery cells shall receive a minimum of two formation discharges before any test cycling. Each cell, during each formation charge, shall be individually charged as necessary to obtain a cell voltage of at least 2.0 volts. The following shall be a part of the formation cycling:

(a) Vacuuming of each cell shall be performed at least:
    (1) After initial filling with electrolyte.
    (2) At the end of the soaking period.
    (3) At the end of each formation charge.
(b) Adjust electrolyte, if required, after each vacuuming.
(c) For each individual cell, the quantities of electrolyte that are used for initial filling and are added or removed from a cell during formation shall be measured and recorded.

4.6.4.2 Resistance test. The internal resistance of each sample and battery cell, in a discharged state after formation, shall be measured between terminals of opposite polarity. The resistance shall be measured with a J. W. Dice Co., model 150 or 153 microohmmeter, Keithley Instrument Inc., model 503-30 milliohmmeter, or equal. Cells shall meet the requirements of 3.3.6.

4.6.4.3 Inspection of wet sample cells. After formation (see 4.6.4.1), each group of sample cells selected in accordance with 4.3.2, shall be series-connected as a 4-cell battery and shall be submitted to the cognizant DCASMA to witness the performance cycle. This cycle shall consist of a full charge in accordance with the manufacturer's service manual instructions and a full discharge at 150 A. Each cell shall, during the discharge, be individually removed from the discharge circuit when the cell reaches 1.34 volts. Data for each cell on voltage, current, time, and electrolyte level at the end of charge and discharge shall be monitored

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

and recorded. The results of the capacity discharge test for each cell shall be averaged and the estimated standard deviation calculated according to the following formula:

$$s = \left(\frac{\sum_{1}^{n} (x_i - \bar{x})^2}{n-1} \right)^{1/2}$$

Where:

s = Estimated standard deviation.
n = Number of sample cells.
$\bar{x}$ = Average of results of test.
x_i = Result of individual sample cell.

The capacity and voltage of each cell shall meet the requirements of 3.3.1 and the average percent of rated capacity for all cells shall be equal to or greater than 100 percent of rated capacity plus 3.1 s. No more than two cycles are permitted to demonstrated rated capacity plus 3.1 s. Adjustment of electrolyte level or individual cell charging during or after the performance cycle shall invalidate that cycle.

4.6.4.4 Notification to ship cells. If notified in writing by NAVSEA the contractor shall be required to ship cells dry, or to fill and ship cells wet, or to fill and form the cells prior to shipment. In the latter case, the contractor shall demonstrate satisfactory performance as specified in 4.6.4.1 through 4.6.4.3.

4.6.4.5 Notification to ship battery. If notified in writing by NAVSEA, the contractor shall be required to ship the battery dry, or to fill and ship the battery wet, or to fill and form the battery prior to shipment. In the latter case, the contractor shall demonstrate satisfactory performance as specified in 4.6.4.1, 4.6.4.2, and 4.6.4.6.

4.6.4.6 Inspection of wet battery. After notification (see 4.6.4.4) and formation (see 4.6.4.1), each battery of 76 series - connected cells shall be submitted to the cognizant DCASMA to witness the performance cycle. This cycle shall consist of a full charge in accordance with the manufacturer's service manual instructions and a full discharge at 150 A. No more than two cycles are permitted to demonstrate 100 percent rated capacity in accordance with the voltage requirements of table I. Electrolyte levels of all cells shall be recorded after discharge and shall be within the levels set by the manufacturer's service manual. Adjustment of electrolyte level or individual cell charging or discharging during or after the performance cycle shall invalidate that cycle.

4.6.4.7 State of battery for delivery. After demonstration of performance, the battery shall be recharged prior to delivery and shipment, unless otherwise directed by NAVSEA. The Ah input and individual cell electrolyte levels shall be recorded after recharge. The input shall be at least 90 percent of the previous output and the electrolyte levels of all cells shall be within the levels set by the manufacturer's service manual.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

4.6.4.8 Mass. After formation (see 4.6.4.1) and the test of 4.6.4.6, the battery shall be completely filled with dielectric oil (see 3.4.17), sealed and weighed. Battery mass shall meet the requirements specified in 3.4.2. If not installed, the mass of the scanner (see 3.3.8) shall be included in the battery mass.

4.7 Shipment of test cells. After completion of the tests on the first article cells (see 4.3.1) and the sample cells (see 4.3.2), all these cells shall be shipped in a discharged state to a Government laboratory as specified (see 6.2.1) to be subjected to life testing as specified in 4.8.

4.8 Life testing. Cells received in a Government laboratory (see 4.7) shall be tested for life over 24 months on several routines as specified and when instructed by NAVSEA to verify conformance with section 3 of this specification.

4.9 Disposition of used cells. Used cells, after retest (see 4.6.3.10), may be submitted to analysis at NAVSEA's discretion prior to shipment to the Government silver reclamation activity as directed by NAVSEA (see 6.2.1).

4.10 Test procedures. The contractor shall prepare a test procedure to cover first article and quality conformance inspections (see 4.2) and a test procedure to cover inspection of wet sample cells and battery (see 4.6.4) in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.11 Test reports.

4.11.1 First article inspection report. The contractor shall prepare a first article inspection (see 4.3.1) report in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.11.2 Sample cell (wet condition) test report. The contractor shall prepare a sample cell (see 4.6.4) test report in accordance with the data ordering document included in the contract or order (see 6.2.2), and shall include the following:

(a) Cell formation data (see 4.6.4.1).
(b) Rated capacity plus 3.1 s (see 4.6.4.3).
(c) Charge cut-off voltage for each cell.
(d) Individual cell electrolyte levels after charge and discharge.

4.11.3 Wet battery or wet cell test report. The contractor shall prepare a wet battery or wet cell (see 4.6.4) test report in accordance with the data ordering document included in the contract or order (see 6.2.2) and shall include the following:

(a) Cell formation data (see 4.6.4.1).
(b) Capacity inputs and outputs.
(c) Cut-off voltages for the battery and end voltages of each cell.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

(d) Average discharge voltage for the battery.
(e) Individual cell electrolyte levels after last discharge and last charge.
(f) Mass.

4.11.4 Silver metal accountability report. A report on accounting of silver metal furnished by the Government and used in the cells delivered shall be prepared in accordance with the data ordering document included in the contract or order (see 6.2.2). The report shall include the amount of silver received by the contractor, the amount per cell for active material and for inactive material (grid and plating), and the total amount used in the cells delivered.

4.11.5 Cable certification report. The contractor shall prepare a cable certification report in accordance with the data ordering document included in the contract or order (see 6.2.2). The report shall include all the information recorded in 4.4.2 and the following:

(a) Cable description.
(b) Contractor's name.
(c) Test equipment description.
(d) Test procedures.
(e) Test results.

4.12 Inspection of packaging. Preservation-packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.6.)

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Cells (see 3.6.1). Cells shall be individually unit protected method III in accordance with MIL-P-116 and packed as specified in 5.2.2.2.

5.1.1.2 Cell markers, accessories, and repair parts. Cell markers, accessories, and repair parts (see 3.6.2 and 3.6.3) shall be segregated and each item in the quantity required for use, except as specified herein, shall be unit protected method III in accordance with MIL-P-116 in individual paperboard folding or set-up cartons or fiberboard boxes conforming to PPP-B-636, domestic grade, except as follows:

(a) Cell jars - One to a fiberboard box.
(b) Cell covers - Ten to a fiberboard box. Each cover shall be wrapped or interleaved with a resilient cushioning material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

(c) Electrolyte - Electrolyte shall be packaged in 3 to 4 liter plastic containers conforming to the level A requirements of PPP-C-300.

5.1.1.3 Batteries. Each complete battery shall be individually unit protected method III in accordance with MIL-P-116 and packed as specified in 5.2.

5.1.2 Level C.

5.1.2.1 Cells and batteries. Cells and batteries shall be packaged as specified in 5.1.1.1 and 5.1.1.3.

5.1.2.2 Cell markers, accessories, and repair parts. Cell markers, accessories, and repair parts shall be segregated and each item, in the quantity required for use, shall be packaged in a manner which will afford protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity for immediate use. The contractor's normal preservation-packaging methods may be used when such meets the requirements of this level.

5.1.2.3 Electrolyte. Electrolyte shall be packaged as specified in 5.1.1.2(c).

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2.1).

5.2.1 Level A. Batteries, cells, accessories and repair parts packaged as specified in 5.1, shall be packed as specified in 5.2.2 except the containers shall be of the overseas type or class.

5.2.2 Level B.

5.2.2.1 Batteries. Each battery shall be individually packed in a wood crate conforming to MIL-C-104, type I, class 1 or 2, style optional, except that the container top shall be designed for reusability by the lag screws or similar fasteners in lieu of nailing.

5.2.2.2 Cell quantity per shipping container. Cells packaged as specified in 5.1 shall be packed in a wood-cleated plywood container designed in accordance with PPP-B-601, domestic type, style optional and the following:

(a) Eight cells shall be packed per container.
(b) Six test cells shall be packed per container.
(c) Containers shall accept either filled (wet) or empty cells, (dry).
(d) Container tops shall be secured by screws in lieu of nailing.
(e) Filled cells shall be placed in a plastic bag to entrap any electrolyte spilled during shipment.
(f) Cells shall be segregated within the container by separators or partitions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

(g) Containers with filled cells shall be modified by the addition of structural numbers to prevent container distortion.
(h) Containers shall have a minimum of two wood skids in accordance with the container specification.
(i) Containers shall be reusable.

5.2.2.3 Cell markers, accessories, and repair parts (less electrolyte). Cell markers, accessories, and repair parts packaged as specified in 5.1 shall be packed in fiberboard boxes conforming to PPP-B-636, class domestic or PPP-B-640, class 1. Alternatively, wood containers conforming to PPP-B-601, domestic type; PPP-B-621, class 1; PPP-B-576, class 1; or PPP-B-585, class II may also be used. Container and style selection shall be at the option of the contractor. Container closure and reinforcing as applicable, shall be in accordance with the applicable container specification or appendix thereto.

5.2.2.4 Electrolyte. Electrolyte, packaged as specified, shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, as specified in Department of Transportation Regulations, Title 49, section 173.260, subpart E, gross mass shall not exceed 30 kg. Alternatively, electrolyte packaged as specified shall be packed in wood containers specified in 5.2.2.3 with suitable padding and separators between the electrolyte carboys and inside faces of the wood containers.

5.2.2.5 Consolidated packing. Items specified in 5.2.2.3, when supplied with a complete battery, may be consolidated within a single shipping container provided with items are segregated by intermediate packing utilizing fiberboard boxes. The container(s) shall be provided with two index lists completely itemizing and identifying the container contents. One copy shall be placed inside the container in a location offering ease of accessibility and removal. One copy shall be attached to the exterior side or end of the container in a protected location which will prevent loss or damage to the list during shipment and storage. Lists shall be enclosed in a waterproof sealed bag or envelope.

5.2.3 Level C.

5.2.3.1 Batteries and cells. Batteries and cells shall be packed as specified in 5.2.2.1 and 5.2.2.2.

5.2.3.2 Cell markers, spare parts, accessories, and electrolyte. Cell markers, spare parts, accessories, and electrolyte, packaged as specified, shall be packed in containers acceptable to the common carrier which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing, or method of shipment shall comply with Uniform Freight or National Motor Freight rules and regulations or other carrier regulations as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required (see 6.2.1) or herein, interior packages (unit and intermediate) and shipping containers shall be marked in accordance with MIL-P-116 and MIL-STD-129 as applicable to the levels of packaging and packing specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

5.3.1 Special markings.

5.3.1.1 Cell containers. Cell containers shall be marked with the following:

(a) "DO NOT DESTROY - REUSABLE."
(b) "FRAGILE" and up-arrows (see MIL-STD-129).
(c) "STORE IN A COOL AREA" (preferably minus 7°C to 4°C) and other storage criteria deemed necessary by the contractor for guidance to the receiving and storing activity for filled and dry cells.
(d) Labelling as required by the Department of Transportation.
(e) Structural and handling markings such as center-of-balance and lift points.

5.4 Technical data (see 3.9, 4.10, and 4.11). When specified (see 6.2.1), preservation-packaging, packing, and marking drawings shall be submitted (see 6.2.2). Drawings are not required when previously submitted and reviewed. Drawings, service manuals, test procedures and reports furnished to the contracting activity shall be prepared for shipment in accordance with level C requirements (see 5.1.2 and 5.2.3).

6. NOTES

6.1 Intended use. Batteries covered by this specification are intended for use in the Deep Submergence Rescue Vehicle (DSRV).

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) When only cells are required (see 3.4.14).
(c) When battery assemblies are required (see 3.4.15).
(d) Address for shipment of the batteries (see 3.6.1).
(e) Address for shipment of the spare cells (see 3.6.1).
(f) Total number of batteries and cells to be delivered (see 3.6.1).
(g) Address for shipment of repair parts and quantity (see 3.6.2).
(h) Address for shipment of accessories, and quantity if accessories are required (see 3.6.3).
(i) If engineering services are required and the maximum number of mandays allowed (see 3.7).
(j) When two terminal or four terminal cover is required (see 3.4.4 and 3.8).
(k) Address for shipment of first article and sample cells to a Government laboratory (see 4.7).
(l) Disposition of used cells (see 4.9).
(m) Level of preservation-packaging and methods required (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

(n) Level of packing and methods required (see 5.2).
(o) Special marking required (see 5.3).
(p) When preservation-packaging, packing, and marking drawings are required (see 5.4).

6.2.2 Data requirements. When this specification is used in a contract which invokes the provision of the "Requirements for Data" of the Defense Acquisition Regulation (DAR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the DAR are not invoked in a contract, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Paragraph	Data requirements	Applicable DID	Option
3.7	Report, contractor engineering and technical services (CETS)	UDI-E-23127	------
3.9.1 and 5.4	Drawings, engineering and associated lists	DI-E-7031	Level 2 Design activity designation - contractor Drawing number - contractor
3.9.2	Technical publications for advanced development programs	DI-M-6157	------
3.9.3	Plan, milestone (type II) for critical supply contractors	UDI-A-23015	------
3.9.3	Report, milestone (type II)	UDI-A-23018	------
3.9.4	Engineering change proposals (ECPs) and requests for deviations and waivers (long form)	DI-E-2037	------
4.10	Test procedures	UDI-T-23732	------
4.11.1	Report, first article inspection	DI-T-4902	------
4.11.2, 4.11.3, and 4.11.4	Reports, test	DI-T-2072	------
4.11.5	Certification data/report	UDI-A-23264	------

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the reqirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Guarantee. The guarantee of the cells when series-connected as a battery shall be as specified in 6.4.1 through 6.4.4.

6.4.1 Nothing in the following guarantee provision shall limit any of the rights afforded to the Government by the provisions of the inspection clause including those provisions relating to latent defects, fraud, or gross mistakes that amount to fraud. This guarantee applies notwithstanding inspection and acceptance by the Government or any provision of the contract concerning the conclusiveness thereof. The contractor guarantees the performance of the battery in terms of service operating conditions (months of service or total Ah output), when the battery is operated and maintained in accordance with the service manual. The battery shall be gauranteed for 18 months or to deliver 40,000 Ah whichever occurs first, before the battery capacity drops to 400 Ah as specified herein, when discharged at 100 A to the cell voltage cutoff of 1.35 volts and subject to the temperature limitations of this specification. The period of the guarantee shall begin:

(a) Date the cells are filled and formed in accordance with the manufacturer's instructions, or
(b) 24 months after acceptance of the sample cell test report in accordance with 4.6.4, whichever occurs first.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

6.4.2 In the event of a failure of the battery or at least five cells, as specified in 6.4.3 the contractor shall:

(a) When battery capacity drops below 500 Ah as specified herein, when discharged at 100 A to the cell voltage cutoff of 1.35 volts, and subject to the temperature limitations of this specification within the first 12 months or 26,000 Ah whichever occurs first, the contractor shall replace the total battery, with a new unused battery manufactured in accordance with this specification at no cost to the Government.

(b) After the period specified in (a) and prior to the end of the guarantee life of 18 months or 40,000 Ah whichever occurs first, the contractor shall indemnify the Government on a pro-rata basis (proportionately) for months of service not obtained or total Ah output not obtained, whichever is smaller. (If P being the purchase price of the battery, N the number of months not obtained, Q the total Ah output not obtained, the smaller of the two numbers $P \frac{N}{18}$ and $P \frac{Q}{40,000}$ shall be used.)

6.4.3 The battery shall be considered failed in the following cases:

(a) When grounds develop which cannot be corrected by the contractor to meet the requirements of the service manual. Corrective action by the contractor shall be at his own expense.

(b) When the battery minimum voltage specified in table I limits battery capacity to a percentage specified in 6.4.2 and cannot be restored by contractor corrective action which will be at his own expense.

(c) When five or more cells fail, with any cell considered failed in the following cases:

(1) The cell is internally shorted.

(2) The cell limits the battery capacity to a percentage specified in 6.4.2 and has to be removed in order to raise the overall battery capacity.

(3) The cell develops leaks through cracks, thus causing grounds in the system.

6.4.4 The same guarantee requirements of life, total Ah output, minimum capacity and indemnification specified in 6.4.1, 6.4.2, and 6.4.3 shall apply to each spare cell individually, whether wet or dry. Nothing in 6.4 diminishes the Government's rights as contained in any other provision of the contract, including but not limited to inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

6.5 Total number of cells to be delivered. The cells to be delivered shall comply with the following:

(a) Six cells for first article inspection for the entire contract (see 4.3.1).
(b) Four cells as samples for quality conformance inspection for each set of two batteries (see 4.3.2).
(c) 152 cells for a set of two batteries (see 3.6.1).
(d) 16 cells as spares for each set of two batteries (see 3.6.1).

6.6 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Changes from previous issue. Asterisks (*) are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6140-N522)

☆U.S. GOVERNMENT PRINTING OFFICE: 1980—603-121/3561

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

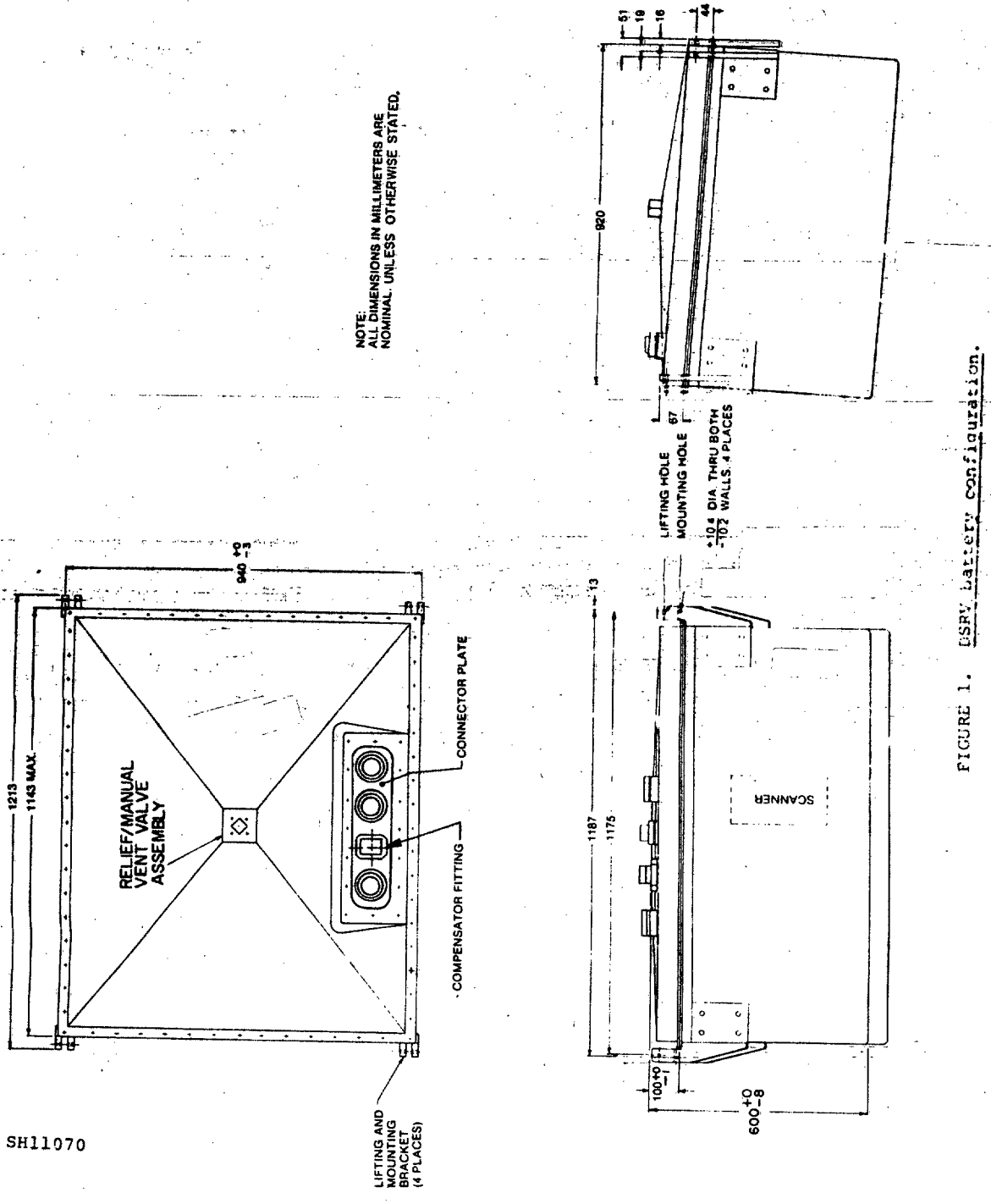
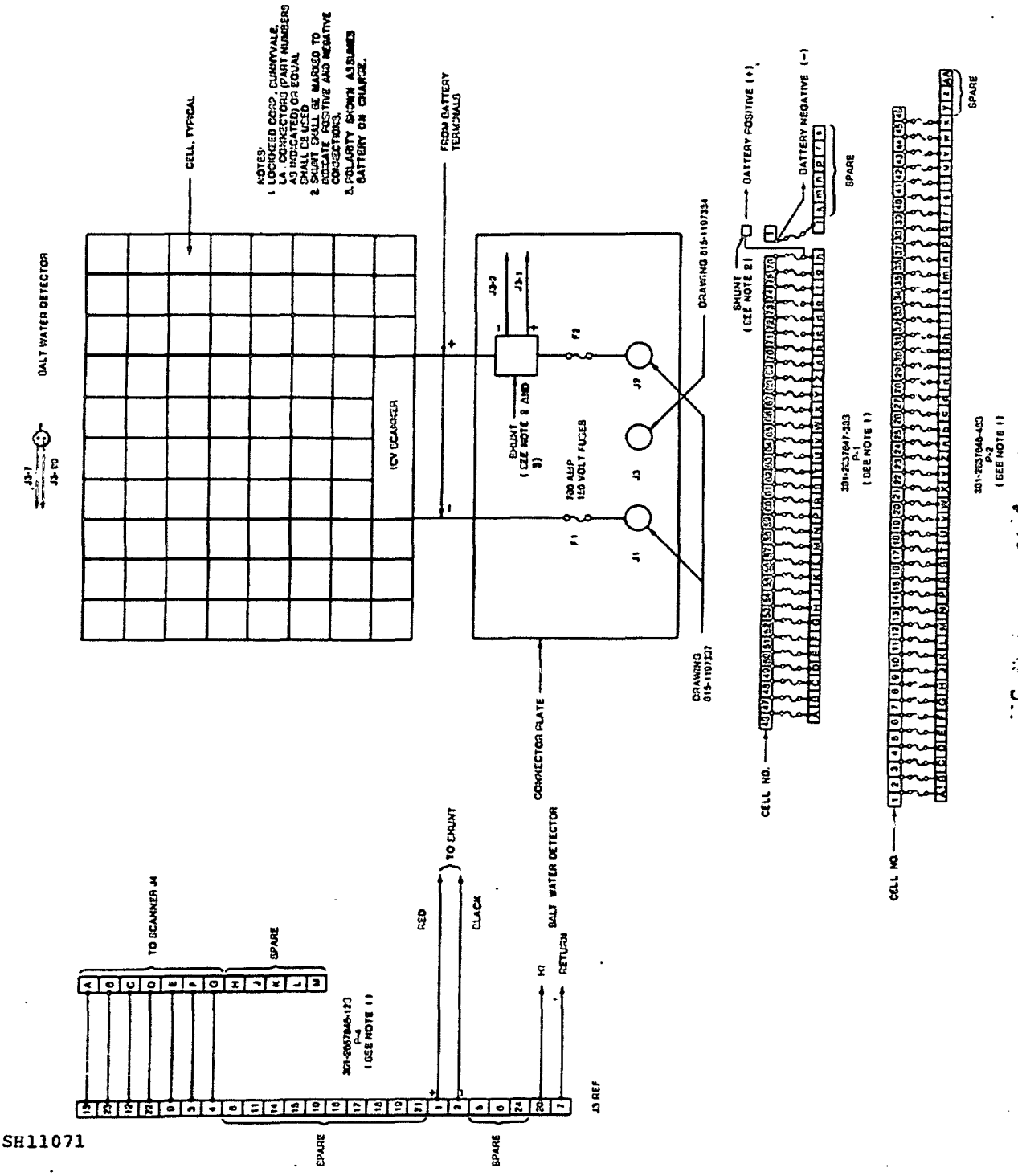


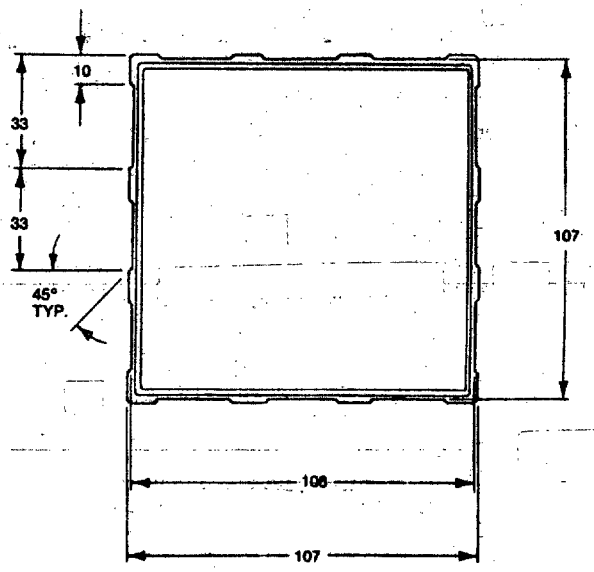
FIGURE 1. ESRV battery configuration.

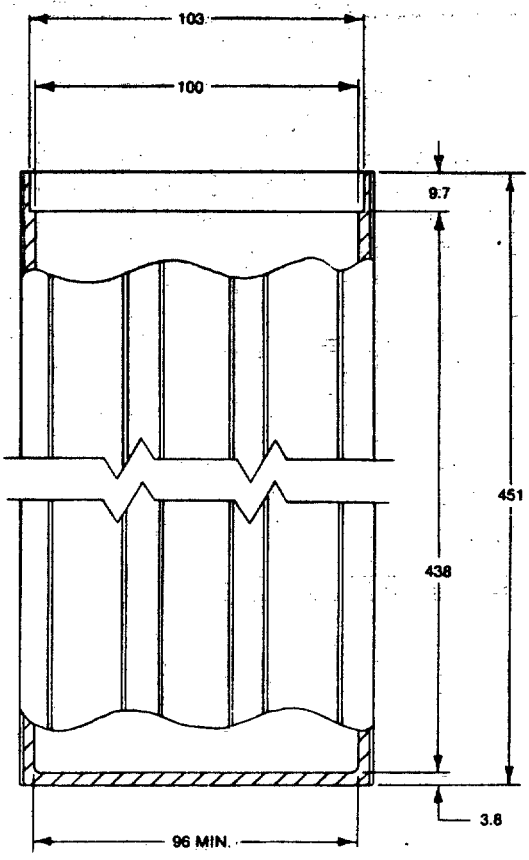
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.



SH11071

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.





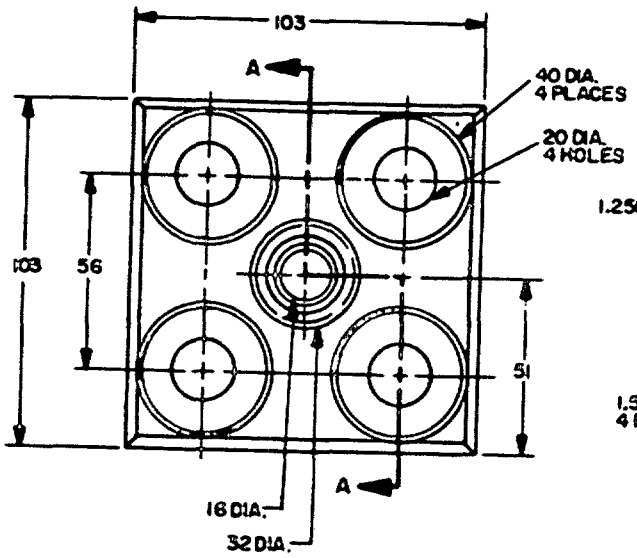
NOTES:

1. ALL DIMENSIONS IN MILLIMETERS ARE NOMINAL.
2. STRESS RELIEVE (ANNEAL) IN AIR AT 165°C ± 3°C FOR AT LEAST 4 HOURS.

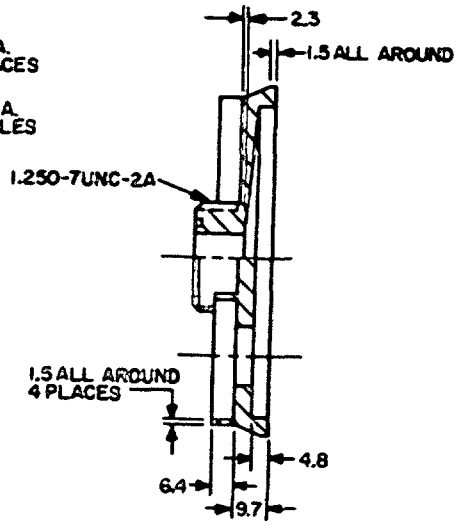
SH11072

FIGURE 3. Cell jar.

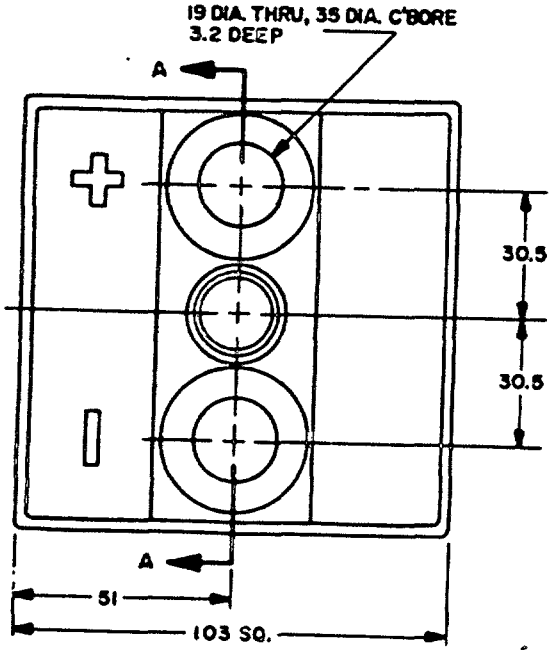
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.



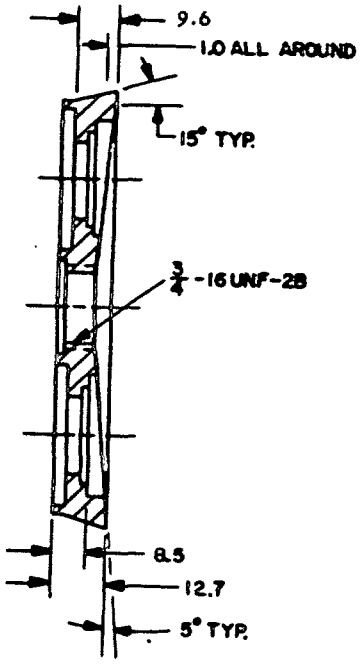
FOUR TERMINALS



SECTION A-A



TWO TERMINALS



SECTION A-A

NOTES:

1. ALL DIMENSIONS IN MILLIMETERS ARE NOMINAL. THREADED HARDWARE DESIGNATION IS UNIFIED THREAD STANDARD.
2. STRESS RELIEVE (ANNEAL) IN AIR AT 165°C±3°C FOR AT LEAST 4 HOURS.

SH11234

FIGURE 4. Cell cover.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

APPENDIX A

POTTING AND ENCAPSULATING WITH POLYURETHANE MOLDING COMPOUND

10. SCOPE

10.1 Scope. This appendix establishes the material and procedures for potting and encapsulating with polyurethane compound (type II) conforming to MIL-M-24041 for use on metal surfaces and neoprene rubber surfaces. This procedure does not apply to materials other than those specified above.

20. APPLICABLE DOCUMENTS

20.1 Not applicable.

30. REQUIREMENTS

30.1 General. The instructions shall be considered as a supplement to the requirements of MIL-M-24041 and shall take precedence only when a specific requirement is identified herein.

30.2 Materials.

30.2.1 Molding compound. The polyurethane compound (hereafter referred to as C) shall be supplied in pre-mixed frozen cartridges. Storage life of pre-mixed frozen cartridges shall be 7 days at minus 30°C, 21 days at minus 40°C and 28 days at minus 62°C. Overage materials shall be discarded.

30.2.2 Metal primer. The metal primer (hereafter referred to as M) shall be used for all metal surfaces. The storage life is 1 year when stored in original containers below 27°C. Material older than 1 year shall be discarded.

30.2.3 Rubber primer. The rubber primer (hereafter referred to as N) shall be used for all neoprene rubber surfaces. The storage life is approximately 3 months when stored in original unopened containers below 27°C. Material older than 3 months shall be discarded (see note in 30.4.2.1).

30.3 Surface preparation. Potting and bonding surfaces shall be prepared in the applicable manner specified in 30.3.1 through 30.3.2.4.

30.3.1 Metal surfaces.

30.3.1.1 Degreasing. Oil, grease, and other contaminants shall be removed by subjecting to vapor degreasing or by thoroughly washing with ethyl alcohol.

NOTE: Good ventilation is required when using ethyl alcohol. There shall be no open flames or sparks near work area. Ethyl alcohol resistant protective gloves shall be worn.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

30.3.1.2 Sandblasting. Surfaces that are not to be sandblasted shall be protected with masking tape or masking jigs. Surfaces that are to be bonded shall be sandblasted with 60 to 100 mesh abrasive garnet. A progressive sandblast with the first blast removing the surface contaminants and a secondary blast using clean dry abrasive garnet to sandblast white metal is required. The sandblast equipment should be supplied with filtered oil and water-free air. Sandblasting does not apply to silver-plated metal parts.

30.3.1.3 Solvent cleaning. Masking tape or masking jig shall be removed and the sandblasted area washed with ethyl alcohol as specified in 30.8. Clean washed rags or lint-free paper towels shall be used.

NOTE: Protective synthetic rubber or plastic gloves shall be worn. Sandblasted surfaces shall not be touched with bare hands or allowed to be contaminated with moisture, oil, or grease.

30.3.1.4 Visual appearance. The sandblasted surface shall be examined for presence of any contamination prior to priming. Where applicable, the matching "O" ring surface of the plug, stuffing tube, etc. shall be examined. Scratches or gouges in this area shall be cause for rejection. Contaminated areas shall be reworked.

30.3.1.5 Priming. Sandblasted surfaces shall be primed within 30 minutes of sandblasting as specified in 30.4.1.

30.3.2 Neoprene rubber surfaces. Surfaces well past areas to be encapsulated shall be cleaned and abraded.

30.3.2.1 Solvent cleaning. Grease, oil, or wax shall be removed by cleaning all potting surfaces with ethyl alcohol (see 30.8) as follows:

(a) Pour or squirt a liberal amount of clean ethyl alcohol over the rubber surfaces.
(b) Wipe off with a clean lint-free cloth or paper towel soaked with clean ethyl alcohol by wiping in one direction only.

30.3.2.2 Surface abrasion. Surfaces shall be abraded with a hacksaw abrading tool (see figure A1) or No. 36 (Bear Brand) grade abrasive paper, or equal. For abrading delicate parts, finer grit abrasive paper shall be used. Surfaces shall be abraded in one direction only. Rubber and abrading particles shall be removed by wiping clean with paper toweling soaked with ethyl alcohol. Cleaned, abraded surfaces shall not be touched with bare hands or allowed to be contaminated with moisture, oil, or grease.

NOTE: Good ventilation is required when using ethyl alcohol. There shall be no open flames or sparks near work area. Ethyl alcohol resistant gloves shall be worn.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

30.3.2.3 Visual appearance. Treated surfaces shall be rough and clean in appearance. Glossy surfaces shall be reabraded and cleaned.

30.3.2.4 Priming. Abraded surfaces with plastic sheets shall be covered until ready to apply primer.

30.4 Priming applications. For safety precautions during priming (see 30.9).

30.4.1 Metal surfaces. Primer shall be applied to sandblasted surfaces within 30 minutes of surface preparation, unless assembly of related components is required.

30.4.1.1 Mixed procedure for primer M. Lot number and date shall be recorded. Primer shall be prepared in accordance with the manufacturer's instructions.

30.4.1.2 Primer application. One brush coat of M shall be applied to the metal surfaces of the sleeve to be encapsulated. The primer should be opaque, approximately 0.05 to 0.07 mm thick. Primer shall dry for 1 hour at room temperature but not over 4 hours.

NOTE: Paramount brush or authorized equivalent only shall be used.

30.4.1.3 Repriming. If surface becomes contaminated or encapsulation has not been done within 4 hours after priming, 30.3.1 and 30.4.1 shall be repeated.

30.4.2 Neoprene rubber surfaces. To obtain good adhesion to neoprene surfaces, the surfaces shall be prepared as specified in 30.3.1.5 prior to priming with N.

30.4.2.1 Priming with N. A small quantity of N primer shall be poured into a disposable container; original container shall be kept closed when not in use. A liberal amount of the primer shall be applied to the clean, abraded, neoprene surfaces by brushing with a "natural horsehair" acid brush and allowed to air dry for approximately 30 minutes at room temperature. After 30 minutes drying time, excess N primer shall be wiped off with a clean, gauze pad and the potting or molding procedure shall begin. Drying time of N shall not exceed 4 hours at room temperature before potting or molding. If primed surface becomes contaminated or potting or molding is not accomplished within 4 hours after application of N, buff neoprene and repeat priming procedure.

NOTE: N is hygroscopic and shall be kept free of moisture. When N hydrolyzes, a dark grainy precipitate is formed, decreasing the primer usefulness. Material containing precipitate shall be discarded.

30.5 Encapsulating procedure. Frozen pre-mixed degassed cartridges of C shall be used in all encapsulation procedures. For safety precautions during encapsulation (see 30.9).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

30.5.1 Storage of frozen cartridges. Upon receipt of shipment, cartridges shall be logged by lot number and date of manufacture. The cartridges shall be stored immediately in a cold facility capable of maintaining temperatures of at least minus 30°C or below. Freezer compartments shall be secured and used for storage of C only. Storage life of the pre-mixed cartridges shall be as specified in 30.2.1.

30.5.2 Thawing of frozen cartridges. Frozen cartridges of C shall be thawed as follows: The cartridge shall be removed from storage and thawed for 40 minutes ± 5 minutes in a heating block[1/] or water bath. Temperature of the compound shall be measured with a thermometer and shall not exceed 38°C. Thawing of the cartridge in an oven shall not be permitted since the time required varies with the oven control and air circulation. When thawing frozen cartridges, cartridges shall be kept in an upright position, nozzle end down with caplug in place, to prevent air from entering and being trapped in the compound.

NOTE: Heating block is the preferred thawing technique. When necessary, the frozen cartridges may be thawed with a water bath in the following manner: A metal container suitable for total immersion of cartridges shall be filled with warm water (38°C) in an air circulating oven. Oven temperature shall be 38°C ± 5°C. Water bath shall be inserted in oven at least 1 hour before thawing cartridges. To maintain proper temperature control, a metal thermometer shall be placed in the water bath. Frozen cartridges shall be inserted into watertight polyethylene bags and placed into water bath for approximately 40 minutes. Make sure that water does not enter into polyethylene bags.

30.5.3 Air displacement. When cartridges have been thawed, entrapped air shall be removed from cartridge by squeezing tube at rear cover and manually pushing cover down until air is removed.

30.5.4 Cartridge insertion. Cartridge shall be inserted into sealant gun. When applicable, end of potting tube (caplug) shall be threaded into fitting that is threaded into molding. Any air in nozzle shall be expelled before attaching gun to mold.

NOTE: Excess moisture and water shall be removed from the C cartridge prior to potting by squeezing out the first few beads onto a paper towel.

30.5.5 Void and air bubble elimination. In applying compound C, every effort should be made to eliminate voids and air bubbles. A small piece of metal, free of contaminants, should be used to tamp or pack compound around wires, irregular surfaces and the like. Tapping on a resilient surface or vibrating the piece being potted will facilitate flow of sealant into the small recesses.

30.5.6 Pot life. The pot life of the thawed cartridges shall be 1 hour at 24°C. Older material shall be discarded.

1/ "Hoedra" temperature controlled tube heater, manufactured by Hoedra, Inc., Monrovia, CA, or equal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

30.5.7 Compound insertion. Potting compound shall be forced into the mold, in one single step without interruption. A two-step phase, with a maximum of 15 minutes interval, is allowed only for air entrapment relief, or for cartridge replacement.

30.5.8 Cure time. The cure time for C shall be a minimum of 16 hours at 82°C. Mold assemblies may be removed after 3 hours at 82°C ± 5°C. When curing requires exposure of a cell to high temperature, the curing temperature shall be 40°C ± 5°C.

30.5.9 Workmanship. The cured C potting compound shall be bubble free, without voids or separations from bonding surfaces. Potting of multiple cables shall be as shown on figure A2.

30.5.9.1 Hardness. The cured C shall have a Shore A hardness of 80 ± 5 when cured at 82°C ± 5°C and 70 ± 5 when cured at 40°C ± 5°C.

30.5.9.2 Adhesion test. After the molded plug has been fully cured, the cable next to the molded portion shall be flexed to see if the molding compound is fully bonded to the cable. Using a blunt probe, the adhesion of the molding compound to the sleeve shall be tested being careful not to damage the molding compound (see figure A3).

30.6 Cleaning. Equipment, tools, and brushes shall be immediately cleaned with ethyl alcohol after use or before material cures. Commercial stripping compounds shall be used to remove cured materials.

30.7 Polyurethane repair.

30.7.1 Imperfections. Small imperfections may be repaired by abrading the surface to remove any mold release or other contaminants, applying more encapsulant, reassembling the mold and curing for another 6 hours at 82°C ± 5°C.

30.7.2 Bubble repair. Bubbles found in the plug may be repaired by inserting a hypodermic into the bubble, attaching a syringe filled with encapsulating material to the needle and forcing the material into the bubble. The needle shall be withdrawn slowly while continuing to force material in until the bubble is filled. If the material cannot be forced in due to trapped air, another hypodermic needle shall be inserted at a different angle to bleed out the trapped air.

30.7.2.1 Curing. Small filled bubbles may be allowed to cure at room temperature; larger bubble repairs will be cured by placing the connector in an 82°C oven for 6 hours.

CAUTION: Do not use solvent for cleaning cured C. Apply new C directly to the buffed surface and cure as recommended. No primer is required.

30.8 Solvent wash procedure. Solvent shall be poured on clean paper towel and the area washed and wiped dry with another paper towel, (i.e., Kimwipes or other untreated toweling). Always wipe away from open end of cable. The towel shall not be dipped in cleaner, but cleaner shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

be poured over the towel. The jacket shall be cleaned well past area to be encapsulated. (Since some part of fittings must be slid back on cable during assembly, it is essential that all this area be clean. Repeat at least three times using clean towels each time. This process is also used on metal parts.

30.9 Safety precautions. The use of any polyurethane identified as containing "CANCER SUSPECT AGENT", shall be limited to a manner approved by OSHA regulations currently in effect. Good general ventilation is adequate for single applications of the process where only a few ounces of primer or solvents are used per hour. In situations where there are high use rates of material, exhaust ventilation systems should be installed. Ethyl alcohol solvent shall not be used near sparks or open flames. If any of the compounds contact the skin, wash vigorously with soap and water. Should any of the compounds get into the eyes, rinse freely with water.

40. QUALITY ASSURANCE PROVISIONS

40.1 General. The following requirements shall be enforced in the control of the characteristics and methods as specified in this procedure.

40.2 Receiving instructions. Each container of C shall be visually examined for the presence of lot number and date of manufacture, contents on the container label, and subsequently stored as specified in 30.5.1.

40.2.1 Sampling. One sample, representative of each lot received, shall be tested for conformance to the requirements of 30.5.9 and 30.5.9.1.

40.2.2 The sample molding batch size should be a minimum of 50 grams and shall be cured at 82°C for 7 hours.

40.3 Production lot testing.

40.3.1 A sample molding of C, upon a piece of copper bus bar (same type as used in production lot), primed with M and prepared as described in 30.3.1, 30.4.1, and 30.5, shall be made with each production molded lot. The molding may be made in a paper or similar container about 20-50 mm high and 50-100 mm in diameter.

40.3.2 A production molded lot shall consist of all units primed at the same time or during the same working day.

40.3.3 A sample molding batch size shall be 25 to 50 grams.

40.3.4 A sample molding batch shall be cured at 82°C for 16 hours and at 40°C for 16 hours.

40.4 Criteria for acceptance.

40.4.1 A production molded lot shall be acceptable if the sample molding reaches Shore A hardness of 80 ± 5 when cured at 82°C ± 5°C or 70 ± 5 when cured at 40°C ± 5°C and is not easily peeled off the primed copper bus bar.

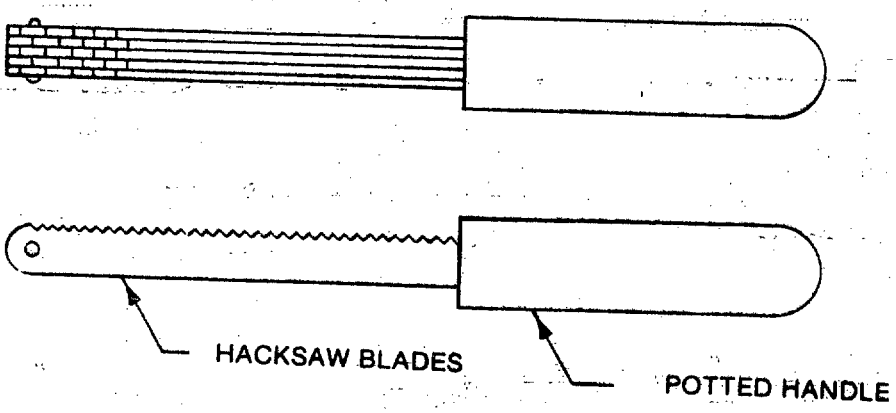
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

40.4.2 If the production molded lot is rejected, and molding material shall be removed and the parts retreated as described in 30.4.1 and 30.5.

40.4.3 Quality control shall perform 100 percent inspection for conformance to the following:

(a) Surface preparation (see 30.3).
(b) Priming applications (see 30.4).
(c) Encapsulation (see 30.5).
(d) Conformance to the requirements of 30.5.9 through 30.5.9.2.

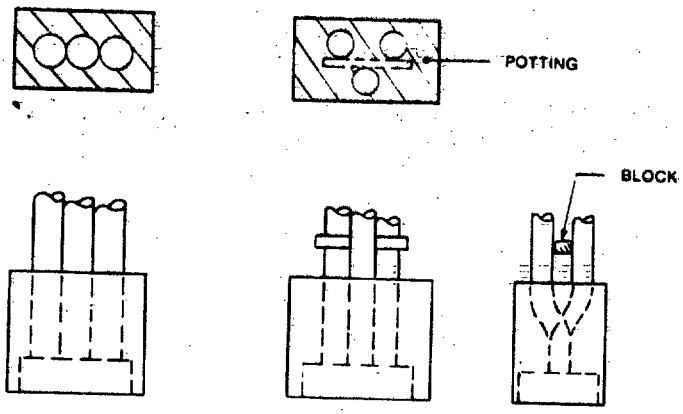
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.



EACH BLADE PLACED IN OPPOSITE DIRECTION TO ONE ALONG SIDE SO THAT TEETH ARE STAGGERED IN OPPOSITE DIRECTION.

SH11074

FIGURE A1. Hacksaw abrading tool.



UNACCEPTABLE

INCORRECT BECAUSE 3 MM MINIMUM GAP BETWEEN CABLES HAS NOT BEEN MET; AS A RESULT NO SEAL AROUND CABLES HAS BEEN ESTABLISHED.

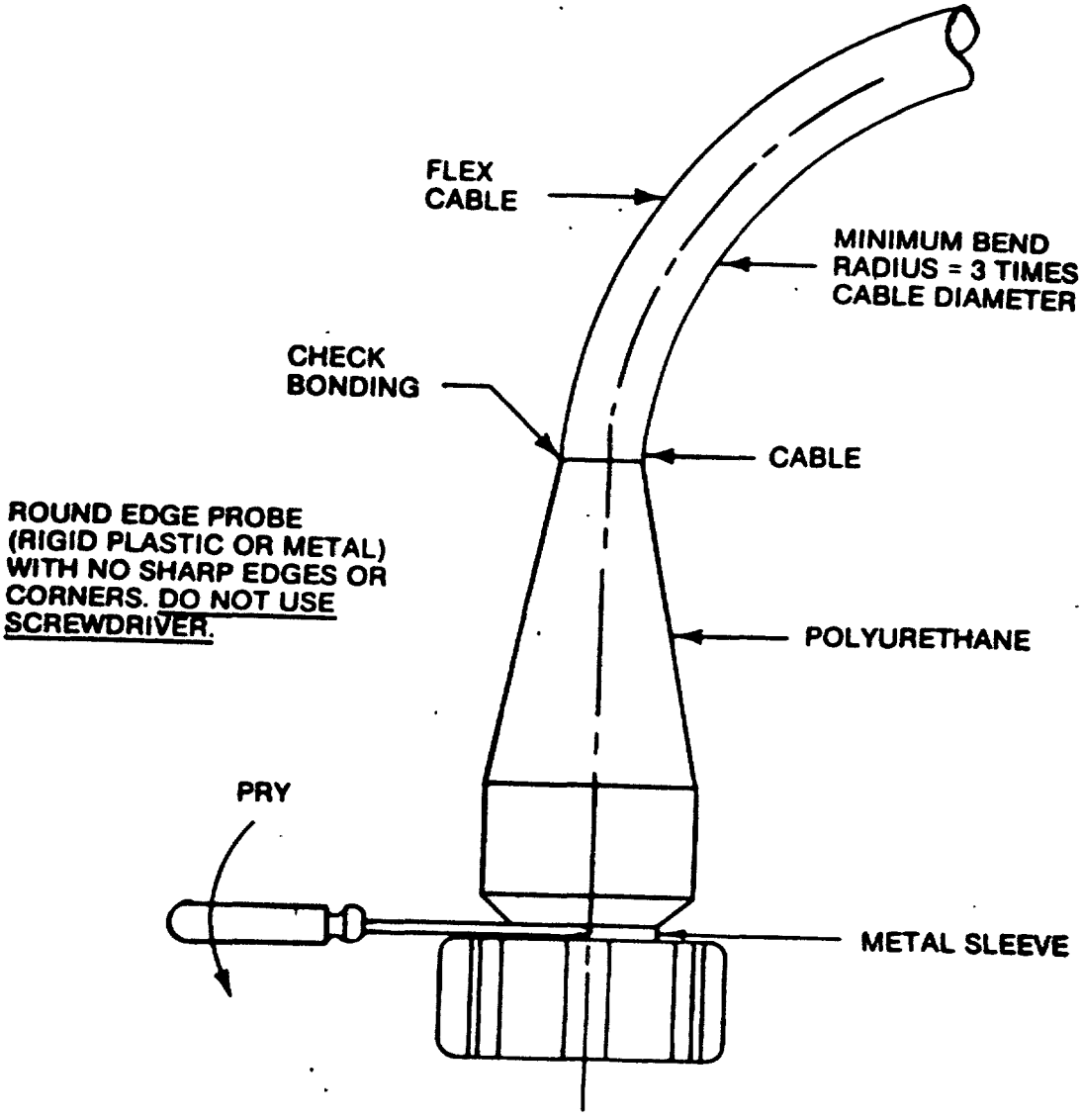
ACCEPTABLE

NOTE 3 MM MINIMUM GAP BETWEEN CABLES OBTAINED BY PLACING BLOCK BETWEEN CABLES THUS SEPARATING THEM DURING POTTING TO ENSURE PROPER ENCAPSULATION.

SH11075

FIGURE A2. Potting of multiple cable systems.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.



SH11076

IF POLYURETHANE MATERIAL PEELS BACK TO REVEAL BARE METAL, THE PLUG ASSEMBLY SHALL BE REJECTED. REPEAT TEST FOR ADHESION TO SLEEVE AT FOUR DIFFERENT POINTS ALONG CIRCUMFERENCE.

FIGURE A3. Adhesion test (nondestructive).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

APPENDIX B

PREPARATION INSTRUCTIONS FOR SILVER-ZINC BATTERY SERVICE MANUAL

10. SCOPE AND OBJECTIVE

10.1 Scope. This appendix covers the instructions which apply to the preparation of a service manual for the silver-zinc type of battery used in Navy submersibles. The manual is provided by the battery manufacturer to describe the required procedures for installation and operation of the specific battery design identified in the manual.

10.2 Objectives. The instructions for the preparation of a battery service manual have several objectives as follows:

(a) Prevent the omission of necessary information by identifying the general subjects to be covered.
(b) Improve the usefulness of the manual by establishing a basic plan for the presentation of the manufacturer's instructions in a fixed sequence under specified general subjects.
(c) Prevent the use of an obsolete issue of a service manual by establishing a method of identifying the issue and date of each manual in a manner easily recognized by the users. Requirement for the preparation of a new service manual (with a new identification number supplied by the Navy) for each new battery contract is also intended to aid identification of the appropriate service manual for each battery design.

20. APPLICABLE DOCUMENTS

20.1 Issues of documents. The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this appendix to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-15071 - Manuals, Technical; Equipment and Systems, Content Requirements for.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

30. REQUIREMENTS

30.1 General. The instructions shall be considered as a supplement to the requirements of MIL-M-15071 and shall take precedence only when a specific requirement is identified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

30.2 Cover. The cover shall be hard cover containing the information shown on example A in accordance with the following and in bold, large characters:

(a) Navy publication number as provided after acceptance of the manual.
(b) Navy designation for the battery.
(c) Navy identification of the ship.

30.3 Title page (first page). The title page shall contain the information as shown in example B in accordance with the following:

(a) Navy publication number as provided after acceptance of the manual.
(b) Navy designation for the battery.
(c) Navy identification of the ship.
(d) Battery manufacturer's name as stated in the contract.
(e) Change number and dates of original and of changes to the manual.

30.4 List of effective pages. The list of effective pages shall be prepared as shown on example C.

30.5 Change record. A permanent change record shall be prepared as shown on example D.

30.6 Approval and acquisition record. An approval and acquisition record page shall be prepared as shown on example E.

30.7 Content assurance. A content assurance record shall be prepared as shown on example F.

30.8 Table of contents. The table of contents shall be organized as shown on example G. The pages shall be consecutively numbered and the page numbers reflected in the table of contents. The titles shall appear as shown and also in the order listed.

30.8.1 Chapters. The manual shall be arranged on the basis of eight chapters.

30.8.2 Section. Certain chapters, as indicated, shall be divided into sections as follows to provide an orderly grouping of related articles:

(a) Sections within each chapter shall be lettered sequentially as A, B, C, etc., prior to the title.
(b) Each section number with title and the corresponding page number shall appear in the table of contents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

30.8.3 Articles. Each chapter shall be subdivided into numbered articles as follows:

(a) Each article shall be identified by a three digit number, with the first digit identical to the chapter number.
(b) Each article shall have a title which is also listed on example G.
(c) Where appropriate, articles may be further subdivided using "a., b., c., and d., etc." for identification prior to the title.

30.8.4 Uniformity. Each manual shall be organized and written to maintain uniformity in the presentation of the information and instructions for the installation and operation of silver-zinc batteries. Wording of the same article shall be identical in all manuals when the intent of the information or the action to be taken is identical for different batteries. If a standard subject for an article as identified on example G is not appropriate for a specific battery design, the text of the service manual for that article shall state "Not Applicable". If a required subject is not included in the standard table of contents, an article with number and title shall be added to the appropriate chapter or section.

30.8.5 Additional text requirements. The following requirements apply to the text of the article indicated:

(a) 212-Cell cover. Include "cover support" and "cover-to-case sealant" discussions.
(b) 217-Flash arresters. Include relationship to electrolyte spillage and cross reference discussion of spill angle limitation in chapter 4, if applicable.
(c) 223-Electrolyte. Identify the KOH concentration and the percentage of impurities present.
(d) 224-Oil. Identify function in pressure compensated cells and the characteristics of oil used.
(e) 245-Hydrogen detector. Identify portable and installed types as appropriate.
(f) 248-Discharge equipment. Identify battery test discharge and shorted cell devices as appropriate.
(g) 262-Voltage. Include specific discussions of voltage on charge, discharge, float, open circuit and the related average, plateau, and limits as appropriate.
(h) 311, 312, and 313-Storage. Identify specific requirements for cells and battery as appropriate.
(i) 321-Safety precautions. This article should cross reference to chapter 8.
(j) 324-Formation. Identify specific requirements applicable to articles 311, 312, and 313 as appropriate.
(k) 326-Oil filling. Include instructions for dye to coloring of oil.
(l) 336-Installation. Procedure should apply to individual cells or assembled battery as appropriate.
(m) 401-Safety precautions. This article should cross reference to chapter 8.
(n) 402-Auxiliary equipment. This article should cross reference to chapter 2, section C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

(o) 403-Jumpered cells. Include instructions for maintenance of good cell disconnected from one battery to balance with jumpered cell in parallel battery as appropriate.
(p) 418-Maintenance charge. Provide instructions as to when to be used.
(q) 420-Voltage limits. Specify the required voltage accuracy.
(r) 424-Charging procedures. Include instructions for parallel batteries as appropriate.
(s) 436-Capacity test discharge. Include instructions for parallel batteries as appropriate.
(t) 504-Maintenance schedule. Include a chart or table with discussion of requirements.
(u) 511-Voltage. Specify the accuracy with which the various cell voltage (charged stand open circuit, charge, discharge and float) can be identified.
(v) 515-Meters. Cross reference to chapter 2, section C.
(w) 519-General cleanliness. Topics should include flash arresters, bubble breakers, cell covers, encapsulating material, battery box (drain and flushing instructions specified) and battery compartment as appropriate.
(x) 521-529-Shipboard action. Instructions in these articles should support the standards identified in articles 511 thorugh 519. Shipboard capabilities may be limited due to battery access for a specific ship system. Required frequency of measurements, particularly for cell voltage shall be clearly stated.
(y) 531-539-Support ship action. Instructions in these articles should also support the standards identified in articles 511 through 519. Support ship requirements shall be identified in conjunction with shipboard capabilities which are directly related to battery access for specific ship system.
(z) 541-General. Include in this article that these recommendations are intended to prolong battery life. Also any action taken based upon these recommendations is subject to not conflicting with ship operations.
(aa) 543-Booster charge. Instructions for use following prolonged open circuit stand as appropriate.
(bb) 544-Temporary storage. Instruction for battery maintenance during extended periods when it is planned not to require battery operation.
(cc) 545-Low temperature. Instructions concerning the advantages of battery storage at low temperatures and the appropriate procedures.
(dd) 546-Electrolyte level. If appropriate, identify the optimum procedure for maintaining the required electrolyte level.
(ee) 547-Characteristics. Characteristics of each battery design should be reviewed to identify additonal recommendations.
(ff) 602-Internal shorting. Identify the characteristics of cells with a "slow short" versus a "hot short".

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

(gg) 612-614-Float, equalizing and maintenance charges. Include discussion of why capacity loss is minimized by each of those charging procedures. Provide cross references to prior discussions in articles 416, 417, and 418 respectively as appropriate.

(hh) 621-622-Detection and verification. Specific instructions in each of these articles related to (1) open circuit charged stand, (2) during recharge, (3) on float, and (4) during discharge as appropriate shall be provided .

(ii) 623-Action. Provide separate instructions for "slow short" and "hot short".

(jj) 624-Troubleshooting diagram. Provide a diagram with appropriate explanation as described in MIL-M-15071. Base diagram on the detection, verification and actions required as appropriate.

(kk) 631-When to replace. Discuss failed cell (shorted) and limiting cell (low capacity).

(ll) 632-How to replace. Cross reference to chapter 7, section 3.

(mm) 701-704-Seals. Discuss the various types of sealing (premixed and prepared by user materials and mechanical seals such as gaskets or "O" rings) as appropriate in each article. Precautions in handling the materials shall be clearly identified.

(nn) 732-734-Replacing cells. Include appropriate article cross references to chapter 3, sections C and D.

(oo) 741-General. Identify possible causes for low capacity and cite cross references to articles in chapter 2, sections D, as appropriate.

(pp) 742-Procedure. Describe the procedure related to each cause discussed in article 741. Cross reference the appropriate article under charging (chapter 4, section B) or cell jumpering (article 403) and cell replacement (chapter 7, section D) as appropriate.

(qq) 801-General. Identify whether any of the recommended safety equipment is supplied by the battery manufacturer.

(rr) 802-809-Equipment. Describe the item, its common name, the manufacturer's name and part number if a particular type is either recommended on the only one available and discuss the purpose. Also provide procedure for use of special devices and cite cross references to other appropriate articles for all items. Article 807 should identify materials such as an eye wash bottle, vinegar, olive oil, Cetapred ointment and clean water. A warning concerning the inadvertent use of the purple K type of fire extinguisher is required.

(ss) 815-Mercury. A mercury warning article shall be included in all manuals.

(tt) 816-Hot short. Include identification and discussion of the hazardous by-products of the "hot short" cell such as smoke, toxic gases, spraying electrolyte, high temperature, etc.

(uu) 817-Sealing materials. Identify possible hazards present during preparation and application and cross reference to supporting informatin in chapter 7, section A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

30.8.6 Titles. Chapter and section titles shall be capital letters and shall have no additional text: Instructions and information shall be associated with a numbered article.

30.8.7 Cross references. Cross referencing of articles of related subjects shall be included to improve the effectiveness of the information presented.

30.9 List of illustrations. The illustrations provided for a specific manual shall be titled and arranged in the order as shown on example G. When a particular illustration is not appropriate the list shall contain that notation instead of the title on the appropriate line as follows:

(a) 5-Cell top connections. Intended to show connections for cooling water, etc., as applicable if not shown in cell assembly (1).
(b) 13-Charge voltage cut-off. Not required if this information is included in the master plan (12).
(c) 16-19-Hydrogen evolution. Effects of temperature and time on evolution to be provided as appropriate.
(d) 20-Effect of temperature. Relationship of initial temperature and charge acceptance to be described.
(e) 21-22-Effect of temperature and pressure. Pressure factor not applicable to all battery designs.

30.10 List of tables. The tables provided for a specific manual shall be titled and arranged in the order shown on example G. When a particular table is not appropriate the list shall contain that notation instead of the title on the appropriate line.

30.11 Appendices. Appendices shall be included in the manual to present instructions and information concerning the operation and use of auxiliary equipment such as the support van charging equipment for the Deep Submergence Rescue Vehicle (DSRV).

30.12 Format. Where appropriate, the format or arrangement of instructions or procedures obtained from another publication shall be used in the original form. Preparation of new text, where required, shall be in a manner similar to the basic organization of the service manual.

30.13 Identification. Appendices shall be identified individually as "A, B, C, etc." and so listed in the manual table of contents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

NAVSEA (a)

SERVICE MANUAL

FOR

SILVER-ZINC BATTERY

(b)

FOR

(c)

Each transmittal of this document outside the agencies of the United States Government must have prior approval of NAVSEA.

DEPARTMENT OF THE NAVY
NAVAL SEA SYSTEMS COMMAND

EXAMPLE A. Cover.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

NAVSEA ___(a)___

SERVICE MANUAL

OPERATING INSTRUCTIONS

FOR

SILVER-ZINC BATTERY

(b) ----------

FOR

(c) ----------

(d) ----------

CONTRACT N00024-XX-X-XXXX

Each transmittal of this document outside the agencies of the U.S. Government must have prior approval of NAVSEA.

DEPARTMENT OF THE NAVY

NAVAL SEA SYSTEMS COMMAND

PUBLICATION RECORD

(e)

ISSUE	DATE
ORIGINAL	----------
CHG. # -----	----------
CHG. # -----	----------
CHG. # -----	----------
CHG. # -----	----------

EXAMPLE B. Title page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

Reproduction for non-military use of the information or illustrations contained in the publication is not permitted. The policy for military use reproduction is established for the Navy in OPNAVINST 5510.1B.

LIST OF EFFECTIVE PAGES

NOTE: On a changed page, the portion of the text affected by the latest change is indicated by a vertical line in the outer margin of the page. Changes to illustrations are indicated by miniature pointing hands.

Total number of pages in this manual is _____ consisting of the following:

Page No.	*Change No.
1-___ ___	0

* Zero in this column indicates an original page.

EXAMPLE C. List of effective pages.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

CHANGE NO.	DATE	TITLE OR BRIEF DESCRIPTION	ENTERED BY

EXAMPLE D. Change record.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

APPROVAL AND ACQUISITION RECORD PAGE

Approval Data For: NAVSEA XXXX-XX-XXX-XXXX

Title of Manual: Service Manual For Silver-Zinc Battery, (enter same data as title page, line 2)

Approval Authority: NAVSEA ltr 5433:XXX:xxx N0024-XX-X-XXXX
Ser XXXX-5433 dtd (enter date of ltr.)

CONTRACT NUMBER	SHIP APPLICABILITY	QUANTITY OF MANUALS	QUANTITY OF EQUIPMENT	INSTALLING ACTIVITY

REMARKS:

DATE_____

CERTIFICATION:

(enter the appropriate certification paragraph from MIL-M-15071 in this space.)

MANUFACTURER'S SIGNATURE
MANUFACTURER'S NAME AND ADDRESS
FEDERAL CODE NUMBER

CHANGE NO.

EXAMPLE E. Approval and acquisition record.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

VALIDATION PERFORMANCE

TITLE OF SERVICE MANUAL FOR SILVER-ZINC BATTERY

CONTRACTOR:

Contract No.

Chapter	Section	Article	Date validation completed	Check here if not validated
1	---	101 through 109		
2	A to E	201 through 268		
3	A to D	301 through 345		
4	A to D	401 through 442		
5	A to E	501 through 547		
6	A to D	601 through 632		
7	A to E	701 through 742		
8	A, B	801 through 817		

Names and authority of validating officer:

Signature of validating officer:

EXAMPLE F. Content assurance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

TABLE OF CONTENTS

Chapter	Section and article	Page
	TITLE PAGE	1
	LIST OF EFFECTIVE PAGES	2
	CHANGE RECORD	3
	APPROVAL AND ACQUISITION RECORD PAGE	4
	VALIDATION PERFORMANCE	5
	TABLE OF CONTENTS	6-
1	GENERAL INFORMATION	
	101 Abreviations and symbols	
	102 Identification	
	103 Description	
	104 Battery	
	105 Cell	
	106 Pressure compensation	
	107 Equipment supplied	
	108 Accessory equipment not supplied	
	109 Guarantee	
2	FUNCTIONAL DESCRIPTION	
	A ELECTROCHEMISTRY	
	201 General	
	202 Reactions	
	B CELL COMPONENTS	
	211 Cell case	
	212 Cell cover	
	213 Cell terminals	
	214 Electrolyte level indicator	
	215 Cooling panels	
	216 Firewalls	
	217 Flash arresters	
	218 Bubble breakers	
	219 Cell marking	
	220 Positive plate	
	221 Negative plate	
	222 Separator materials	
	223 Electrolyte	
	224 Oil	
	C BATTERY COMPONENTS	
	231 Battery box	
	232 Battery cover	
	233 Flushing valve	
	234 Temperature	

EXAMPLE G. Table of contents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

TABLE OF CONTENTS

Chapter	Section and article		Page
2 (con.)	235	Battery connector	
	236	Intercell connectors	
	D	AUXILIARY EQUIPMENT	
	241	Cell voltage monitor	
	242	Voltmeters	
	243	Ammeters	
	244	Ampere-hour meter	
	245	Hydrogen detector	
	246	Ground detector	
	247	Thermometers	
	248	Discharge equipment	
	E	MECHANISMS	
	261	Capacity	
	262	Voltage	
	263	Electrolyte level change	
	264	Heat generation	
	265	Gas evolution	
	266	Cell reversal	
	267	Capacity loss	
	268	Internal cell shorting	
3	PREPARATION AND INSTALLATION		
	A	SHIPMENT	
	301	Dry shipment	
	302	Wet shipment	
	B	STORAGE	
	311	Dry	
	312	Wet charged	
	313	Wet discharged	
	C	PREPARATION FOR INSTALLATION	
	321	Safety precautions	
	322	Filling cells	
	323	Assembling battery	
	324	Formation	
	325	Encapsulation	
	326	Oil filling	

EXAMPLE G. Table of contents (con.).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

TABLE OF CONTENTS

Chapter	Section and article		Page
3 (con.)	D	INSTALLATION	
		331 Parts list	
		332 Shipping list	
		333 Installation hardware list	
		334 Maintenance parts list	
		335 Special tools list	
		336 Installation	
		337 Wedging	
		338 Intercell connectors	
		339 Connect battery terminals	
		340 Connect voltage monitor leads	
		341 Connect water cooling system	
		342 Install level indicators	
		343 Install temperature sensors	
		344 Placing battery in service	
		345 Testing	
4	OPERATION		
	A	GENERAL	
		401 Safety precautions	
		402 Auxiliary equipment	
		403 Jumpered cells	
		404 Parallel batteries	
	B	CHARGING	
		411 General characteristics	
		412 State of charge	
		413 Normal charge	
		414 Overcharge	
		415 Partial charge	
		416 Float charge	
		417 Equalizing charge	
		418 Maintenance charge	
		419 Charge rate	
		420 Voltage limits	
		421 When to charge	
		422 Ventilation	
		423 Temperature limits	
		424 Charging procedures	

EXAMPLE G. Table of contents (con.).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

TABLE OF CONTENTS

Chapter	Section and article		Page
4 (con.)	C	DISCHARGING	
		431 General characteristics	
		432 Discharge rate	
		433 Voltage limits	
		434 Ventilation	
		435 Temperature limits	
		436 Capacity test discharge	
	D	BATTERY RECORDS	
		441 Battery record book	
		442 Battery monthly report	
5		MAINTENANCE	
	A	REQUIRED ACTION	
		501 General	
		502 Capacity maintenance	
		503 Preventive maintenance	
		504 Maintenance schedule	
	B	STANDARDS	
		511 Voltage	
		512 Capacity	
		513 Electrolyte level	
		514 Ground resistance	
		515 Meters	
		516 Ventilation system	
		517 Cell seals	
		518 Battery seals	
		519 General cleanliness	
	C	SHIPBOARD ACTION	
		521 Voltage	
		522 Capacity	
		523 Electrolyte level	
		524 Ground resistance	
		525 Meters	
		526 Ventilation	
		527 Cell seals	
		528 Battery seals	
		529 General cleanliness	

EXAMPLE G. Table of contents (con.).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

TABLE OF CONTENTS

Chapter	Section and article		Page
5 (con.)	D	SUPPORT SHIP AND SHORE BASED ACTION	
		531 Voltage	
		532 Capacity	
		533 Electrolyte level	
		534 Ground resistance	
		535 Meters	
		536 Ventilation	
		537 Cell seals	
		538 Battery seals	
		539 General cleanliness	
	E	RECOMMENDATIONS	
		541 General	
		542 Open circuit versus float	
		543 Booster charge	
		544 Temporary storage	
		545 Low temperature storage	
		546 Electrolyte level	
		547 Additional comments	
6	TROUBLE SHOOTING		
	A	FAILURE MODES	
		601 Capacity loss	
		602 Internal shorting	
	B	CAPACITY LOSS	
		611 Causes	
		612 Float charging	
		613 Equalizing charge	
		614 Maintenance charge	
	C	INTERNAL SHORTING	
		621 Detection	
		622 Verification	
		623 Action	
		624 Trouble shooting diagram	
	D	CELL REPLACEMENT	
		631 When to replace	
		632 How to replace	

EXAMPLE G. Table of contents (con.).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

TABLE OF CONTENTS

Chapter	Section and article	Page
7	REPAIR	
	A SEALS	
	701 General	
	702 Materials	
	703 Preparation	
	704 Application	
	B FLASH ARRESTER	
	711 Cleaning	
	712 Repair	
	713 Replacement	
	C REDUCING GROUNDS	
	721 General	
	722 Action	
	D REPLACING CELLS	
	731 Removing old cell	
	732 Preparing new cell	
	733 Installation	
	734 Battery reassembly	
	E RESTORING BATTERY CAPACITY	
	741 General	
	742 Procedure	
8	SAFETY PRECAUTIONS	
	A SAFETY EQUIPMENT	
	801 General	
	802 Fire extinguisher	
	803 Face shield or goggles	
	804 Neoprene sheeting	
	805 Insulated tools	
	806 Vaseline or petrolatum	
	807 First aid materials	
	808 Glass beaker or large test tube	
	809 Rags or towelling	

EXAMPLE G. Table of contents (con.).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

TABLE OF CONTENTS

Chapter	Section and article		Page
8 (con.)	B	HAZARDS	
		811 Electrical	
		812 Hydrogen	
		813 Electrolyte	
		814 Seawater	
		815 Mercury	
		816 Hot short	
		817 Sealing and encapsulating materials	
		LIST OF ILLUSTRATIONS	
		LIST OF TABLES	
		USER ACTIVITY COMMENT SHEET	

EXAMPLE G. Table of contents (con.).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

LIST OF ILLUSTRATIONS

Number	Title	Page
1	Cell assembly	
2	Battery box	
3	Battery assembly	
4	Battery cell layout	
5	Cell top connections	
6	Battery cell interconnections	
7	Receptacle plate assembly	
8	Wiring diagram	
9	Level indicator and interpretation	
10	Temperature sensor assembly	
11	Electrolyte adjustment device	
12	Master plan, curves and data	
13	Charge voltage cut-off	
14	Typical charge curves	
15	Typical discharge curves	
16	Hydrogen evolution on stand	
17	Hydrogen evolution on float	
18	Hydrogen evolution on charge	
19	Hydrogen evolution on discharge	
20	Effect of temperature on charge	
21	Effect of temperature and pressure on discharge	
22	Effect of temperature and pressure on charge	

EXAMPLE G. Table of contents (con.).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

LIST OF TABLES

Number	Title	Page
1	Equipment supplied	
2	Other equipment required	
3	Electrolyte specification	
4	First aid materials	
5	Scheduled maintenance	
6	Installation kit	
7	Service kit repair parts	
8	Oil specification	

EXAMPLE G. Table of contents (con.).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
DOD-B-24505C(SH)METRIC

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T22:15Z
Check the source to verify that this is the current version before use.